tsitski ProBuilder Tools

UNITY ASSET | PROBUILDER 5+ | UNITY 2021.3+

A live boolean cutter, level generators and modelling tools for Unity ProBuilder.

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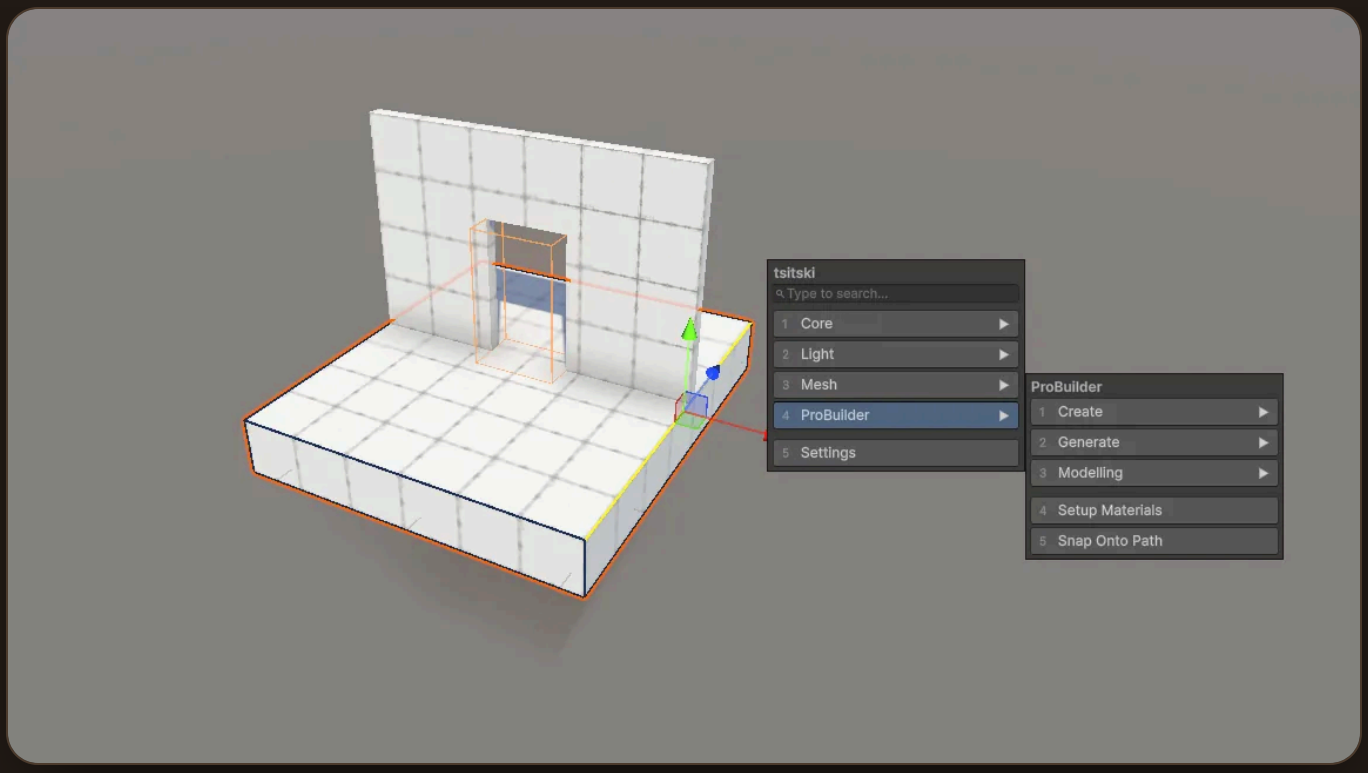
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Overview



tsitski ProBuilder Tools adds live, non-destructive building blocks for level design in Unity ProBuilder: a boolean cutter for doors and windows, generators for polygons, roofs, trims, arrays, stairs, pillars and walls, and a set of modelling tools that are one shortcut away in the Scene view.



Doorways and windows cut live into a ProBuilder wall.

Components

- **Array** — copies a ProBuilder mesh, a pillar or a prefab along a path, evenly spaced or split into divisions.
- **ProBuilder Cutter** — a box, rounded box, cylinder, polygon, explosion, path or mesh that cuts a hole through any ProBuilder mesh it overlaps, live in the editor.
- **ProBuilder Trim** — sweeps a 2D profile along a path for skirting, mouldings, pipes and railings.
- **ProBuilder Stairs** — stairs along a path or around a spiral, with landings, winding corners, an optional stairwell cutter, side trims and a smooth collider.
- **ProBuilder Pillar** — a base, shaft and capital spun into a round or square pillar, or swept around a wall corner, snapping onto paths.
- **ProBuilder Floor Plan** — walls along a path or a floor's outline, with trims, arrays and a roof generated on top.
- **ProBuilder Polygon** — one or more outlines merged and extruded into a slab, with holes that can overlap or cross the edge.
- **ProBuilder Roof** — hip, valley and gable roofs from the center lines of a building's wings, each with its own width and height.

- **ProBuilder Path** — raised strips along curved branches for sidewalks, kerbs and paths, with a width and height at every point and junctions wherever branches meet.

Modelling tools

- **Bevel** and **Inset Faces** — with a live settings window, like Blender's redo panel.
- **Bridge Faces** — join two groups of faces with a tube or tunnel.
- **Collapse Edges or Vertices** — merge flat faces and clean up leftover vertices.
- **Split Path**, **Center Transform To Mesh**, **Outline Trim** and **Snap Onto Path**.
- **Placement** — snap objects to the grid or onto walls, and drop them with physics.
- **Generate Lightmap UVs** — rebuild lightmap UVs on every generated mesh at once.
- **Setup Materials** — paint whole buildings at once, sorting faces into inside and outside walls, floors, roofs and trims.
- **Tools popup** — press `Alt` + `X` in the Scene view to get every tool under your cursor, with search and number keys.

Installation

1. Buy or add the package on the Asset Store, then open **Window** › **Package Manager**, choose **My Assets**, and import **tsitski ProBuilder Tools**.
2. If ProBuilder isn't installed yet, a prompt offers to install it (also under **Tools** › **tsitski** › **ProBuilder** › **Install ProBuilder**). Package Manager picks the newest ProBuilder that works with your Unity version.
3. Everything lands in `Assets/tsitski/tsitski ProBuilder Tools`, with shared code and the settings in `Assets/tsitski/tsitski Core`. Open the demo scene in the `Demo` folder to see each tool in action.

NOTE

Requires Unity 2021.3 or newer and ProBuilder 5.0.3 or newer, including ProBuilder 6 on Unity 6. Works with the Built-in, URP and HDRP render pipelines.

NOTE

Updating from 1.4? Polygons are upgraded to the new list of outlines when they load, so save your scenes and Polygon Stats to keep the new format. Stairs **Top Material** is now **Top Step Material** and **Bottom Material** is now **Side Material**; the new **Forward Step Material** starts empty. Roofs get a **Smoothing Angle** of 30, so curved roofs shade smooth on their next rebuild (set it to 0 to keep hard edges). Center Transform To Mesh moved to **Tools › tsitski › Core**. Scripts reading `polygon.path` should use `paths`.

NOTE

Updating from 1.3? The ProBuilder tools moved under **Tools › tsitski › ProBuilder**, and the shared tools under **Tools › tsitski › Core**. **Split Into New Trim** is now Split Path and **Regenerate Lightmap UVs** is now Generate Lightmap UVs. Shortcuts you set on the old menu items need to be set again.

NOTE

Updating from 1.2? The menu moved from **Tools › tsitski ProBuilder Tools** to **Tools › tsitski**, and the components are now called **tsitski ProBuilder ...** in Add Component. Your scenes and Stats assets load unchanged. Scripts that reference the classes need the new names (`ProBuilderTrim` is now `TsitskiProBuilderTrim`, and `ProBuilderArray` is now `TsitskiArray` in `Tsitski.Core`).

TIP

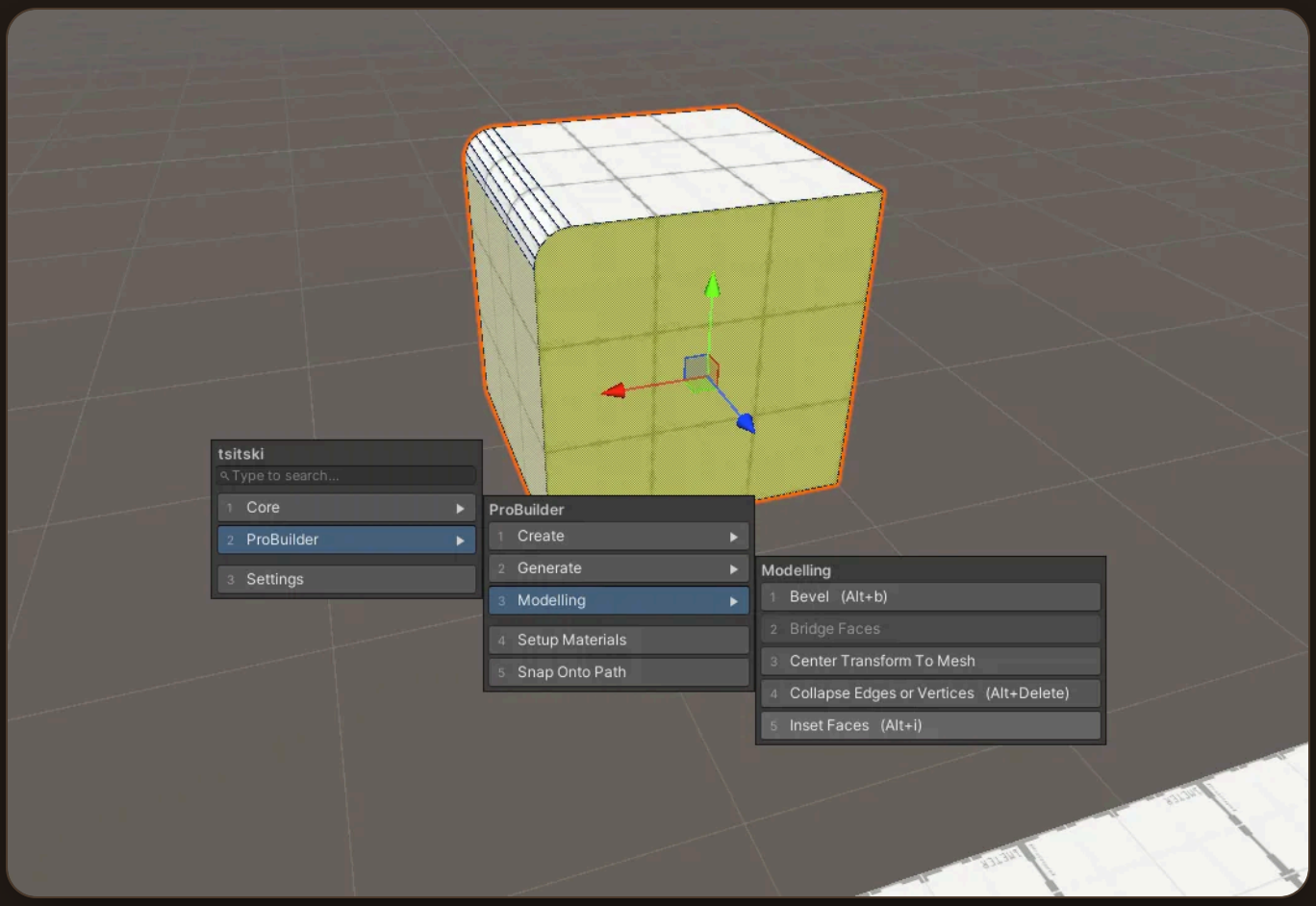
If the demo scene's text shows up pink or Unity asks to import **TMP Essentials**, click **Import TMP Essentials**. It's a one-time setup for TextMeshPro.

Tools menu and popup

Every tool lives in `Tools` › `tsitski`. **Core** holds the tools shared by every tsitski package (Placement, Split Path and Generate Lightmap UVs), **ProBuilder** holds **Modelling**, **Generate**, **Setup Materials** and **Snap Onto Path**, and **Settings** sits at the bottom. Each group starts with a **Create** menu that adds a new generator at the Scene view's pivot, also found under `GameObject` › `tsitski ProBuilder Tools`. Other tsitski packages, like tsitski Mesh Tools, add their own group.

Faster: hover the Scene view and press `Alt` + `X` to open the same menu right under your cursor. Hover a group to open it to the side. Tools that can't run on your current selection are greyed out.

- **Number keys** — `1` to `0` press the matching row, opening groups and running tools. `Backspace` goes back a level.
- **Search** — start typing to search every tool. The closest matches are listed at the top; press `Enter` to run the first one or a number to pick one.
- **Closing** — `Esc`, `Alt` + `X` again, or clicking in the Scene view or another window.



Alt + X opens every tool at your cursor.

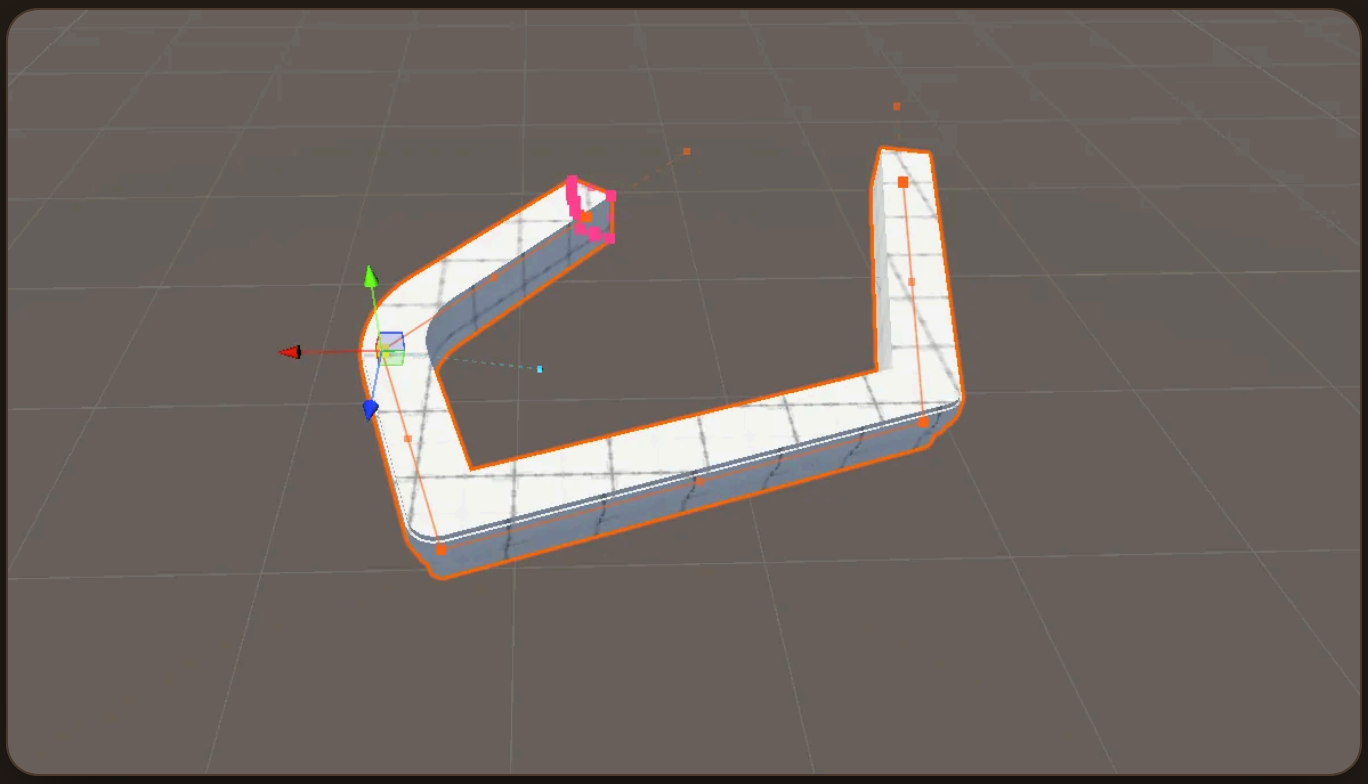
PROPERTY	SHORTCUT	WHAT IT DOES
Tools popup	Alt + X	Open every tool at the cursor.
Bevel	Alt + B	Bevel the selected edges or faces.
Inset Faces	Alt + I	Inset the selected faces.
Collapse	Alt + Delete	Collapse the selected faces, edges or vertices.

TIP

Change any shortcut in `Edit` › `Shortcuts`. The popup is under **tsitski › Tools Popup**.

Editing points and paths

Cutters, polygons, roofs, trims, arrays, stairs, pillars, floor plans and paths all edit their points the same way, right in the Scene view. Select the object and its points appear.



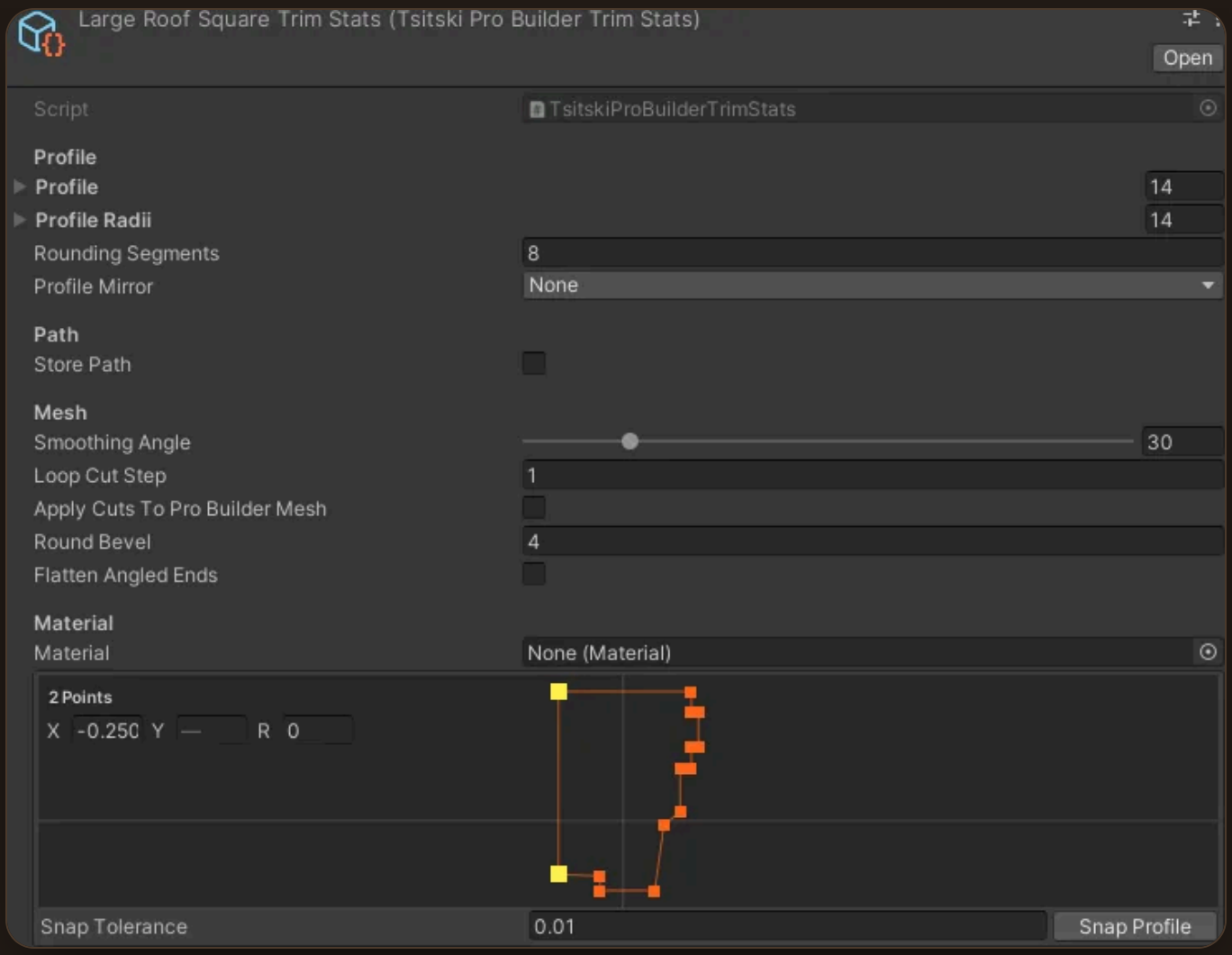
Points, edge-midpoint dots, extend dots and the cyan rounding handle.

- Click a point to select it. `Shift` + click adds or removes points, and `Shift` + drag on empty space box-selects them. `Shift` + click on another object still adds it to the selection as normal.
- Selected points follow the current tool: Move (`W`), Rotate (`E`) or Scale (`R`), around the middle of the selection. On a Path cutter, rotating also tilts the cutter's box at each selected point, so even a single point can be rotated.
- Click the small dot in the middle of an edge to insert a point there.
- On an open path, click the dot past either end to extend the path.
- Drag the cyan handle on a selected corner to round it. Rounding is saved per point, and the object's **Rounding Segments** (or **Corner Segments**) sets how smooth it is.
- `Delete` removes the selected points and `Esc` clears the selection. On objects with several paths (polygon outlines and holes, roof wings and path branches), deleting every point of a path, or leaving it with too few points, removes the whole path.
- `Ctrl` + `C` and `Ctrl` + `V` (`Cmd` on Mac) copy and paste points, even between objects. Pasted points start right after the copied ones; as you drag them they swap to before the copied point if that keeps the path tidier.
- Copy a whole path (every point selected) on a polygon, roof or path and paste it to add it as a new outline, hole, wing or branch, selected and ready to drag into place.

Points are edited only when exactly one object is selected. Generated meshes are rebuilt from their settings, so ProBuilder edits made directly to their shape are overwritten.

Stats assets

Cutters, polygons, roofs, trims, arrays, stairs, pillars, floor plans and paths each have a matching **Stats** asset (create one with `Assets` › `Create` › `tsitski ProBuilder Tools`, Array Stats under `Assets` › `Create` › `tsitski`, or the inspector's **Save As ... Stats** button). Assign it to share settings between many objects: editing the asset updates every object that uses it.



Settings held by the Stats asset are hidden on the component.

- While a Stats asset is assigned, the settings it holds are hidden on the component.
- **Copy Stats Into ...** copies the asset's settings back onto the component and unassigns it, so you can tweak one object on its own.
- Some settings are often different per object, so the Stats asset can leave them out: **Store Height** and **Store Thickness** on Floor Plan Stats, and **Store Height** and **Store Width** on Path Stats. Turn one off and each object keeps its own value.
- Every inspector also has a **Rebuild** button.

> TIP
> Hover any field in an inspector for a tooltip explaining it.

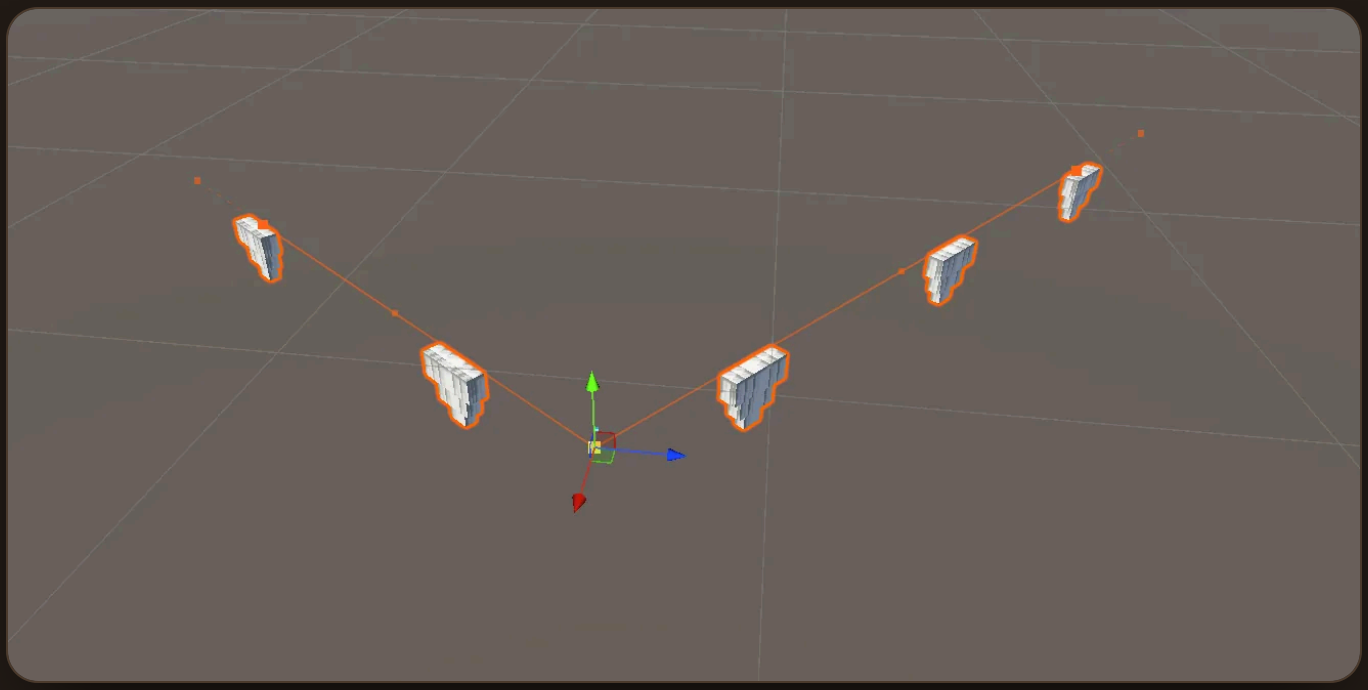
Materials

Generated meshes get their materials in three ways, and all three survive rebuilds.

- **Locked materials.** Trims, arrays, pillars and roofs have a **Material** field, polygons have **Top** and **Bottom Material**, stairs have **Top Step**, **Forward Step** and **Side Material**, paths have **Top**, **Side** and **Bottom Material**, and floor plans have **Inside** and **Outside Material** for the two faces of their walls. These can also live in the Stats asset. A locked material is reapplied on every rebuild and Setup Materials leaves those meshes alone. Leave a slot empty to keep those faces free.
- **Painting faces by hand.** Faces you give a material with ProBuilder's Material Editor or by dragging a material onto them keep it when the mesh rebuilds, as long as they're still in roughly the same place.
- **Default Mesh Material.** Anything that has neither gets the default material from Settings.

Array

An array copies a piece along a path: fence posts, columns, lights, brackets, railings. The piece can be a ProBuilder mesh, ProBuilder Pillar Stats, or a prefab or model.



An array of a source mesh along a path.

1. Add `Add Component` › `tsitski` › `tsitski Array` to an empty GameObject.
2. Assign a **Piece**, or click **Create Source** to make a small ProBuilder cube to model. The piece's +Z points along the path and +Y points up.
3. Edit the path in the Scene view.

Pieces

- **ProBuilder mesh** — copied into the array's own combined mesh. The inspector has **Select Source**, **Rotate Source 90° Y**, and **Save / Load Source** to move the mesh in and out of the stats.
- **ProBuilder Pillar Stats** — a pillar at every slot, **Pillar Height** tall. Corner pillars snap onto the path's corners and face the side set by **Corner Side**. Turn on **Merge Into Mesh** to combine them into the array's mesh instead of separate objects.
- **Prefab or model** — placed as separate objects, or merged with **Merge Into Mesh**.

Spacing

- **Split At Corners** treats each straight run between corners on its own, so spacing restarts at every corner and corners always get a copy.
- **Division Length** splits every run into equal divisions and spaces copies inside each one. Match it to another array's spacing to fill the gaps between its copies, and use **Skip Divisions** so the two don't overlap.
- **Fit Spacing** stretches the spacing so copies land evenly from end to end; off keeps the exact **Spacing**, starting at **Offset**.
- **Placement › Span Ends** places, at every slot, one copy facing into each span next to it, for example brackets on both sides of a post.

PROPERTY	TYPE	WHAT IT DOES
Stats	Array Stats	Shared settings. It can also hold the piece, the source mesh itself (**Store Source**) and the path (**Store Path**).
Material	Material	Locked onto every face of the generated mesh. See Materials.
Piece	Object	What's copied: a ProBuilder mesh, Pillar Stats, or a prefab or model.
Pillar Height	float	Height of each pillar when the piece is Pillar Stats.
Merge Into Mesh	bool	Combine pillars or prefabs into the array's mesh instead of separate objects.
Corner Side	enum	Outside or Inside: which side of the path corner pillars sit on.
Placement	enum	Spacing places one copy per slot. Span Ends places one facing into each span.
Piece Rotation	Vector3	Extra rotation for every copy, for pieces modelled facing another way.
Spacing	float	Distance between copies.
Offset	float	Moves the first copy along each division or run when Fit Spacing is off.
Fit Spacing	bool	Adjust the spacing so copies land evenly from end to end.
Split At Corners	bool	Space each straight run on its own, with a copy on every corner.
Division Length	float	Split runs into equal divisions about this long. 0 is off.
Skip Corners	bool	Leave out the copies on corners and the ends of an open path.
Skip Divisions	bool	Leave out the copies on division points.
Align To Path	bool	Turn each copy to follow the path.
Rounding Segments	int	How smooth rounded path corners are.

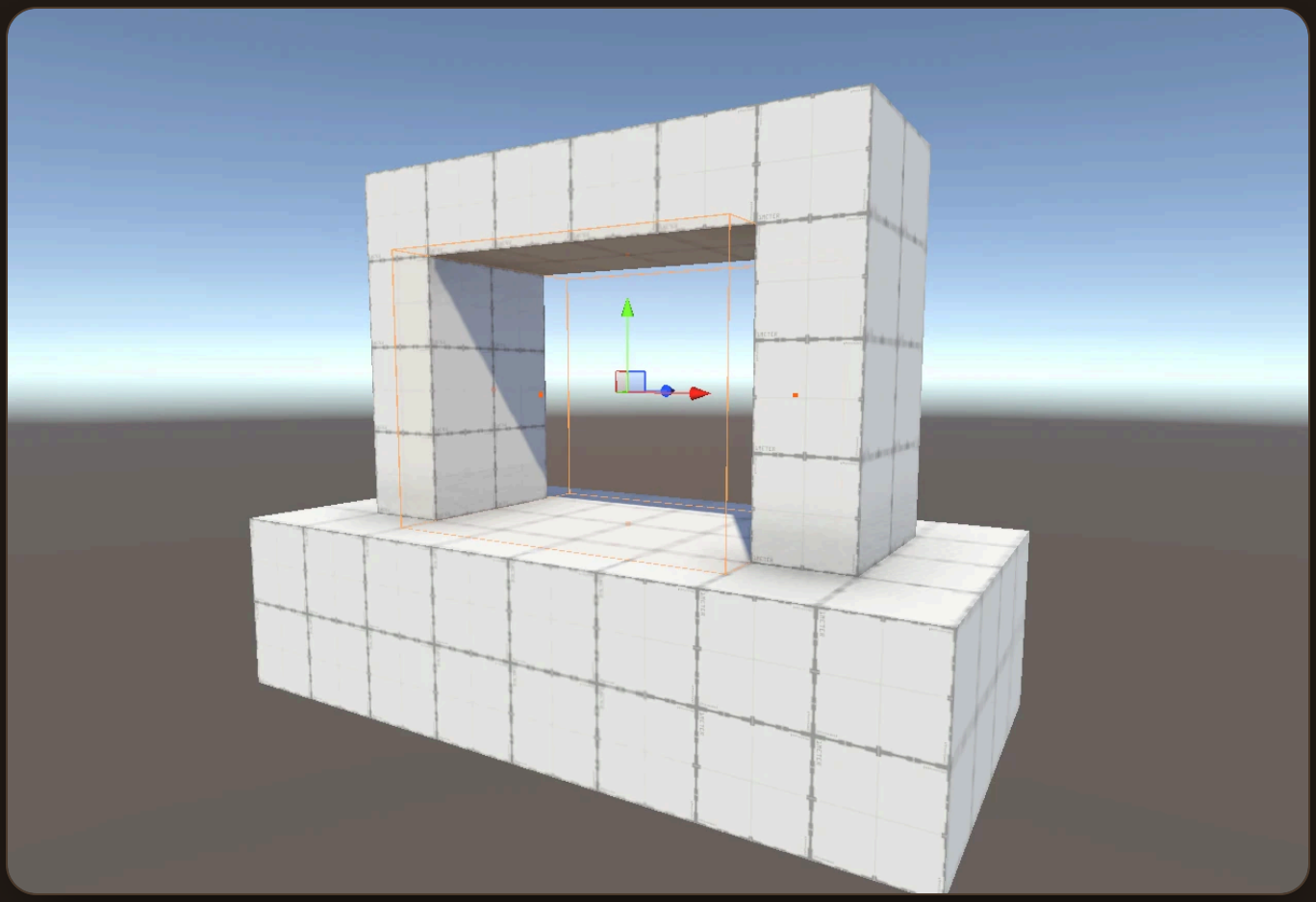
PROPERTY	TYPE	WHAT IT DOES
Closed	bool	Joins the last point back to the first.

TIP

A Floor Plan layer with Array Stats whose piece is corner Pillar Stats puts a pillar in every corner of the building. See ProBuilder Floor Plan.

ProBuilder Cutter

A ProBuilder Cutter is a solid shape. Wherever it overlaps a ProBuilder mesh, that part of the mesh is cut away. The cut updates live as you move, scale or edit the cutter or the mesh, so you can slide a doorway along a wall and watch it follow.



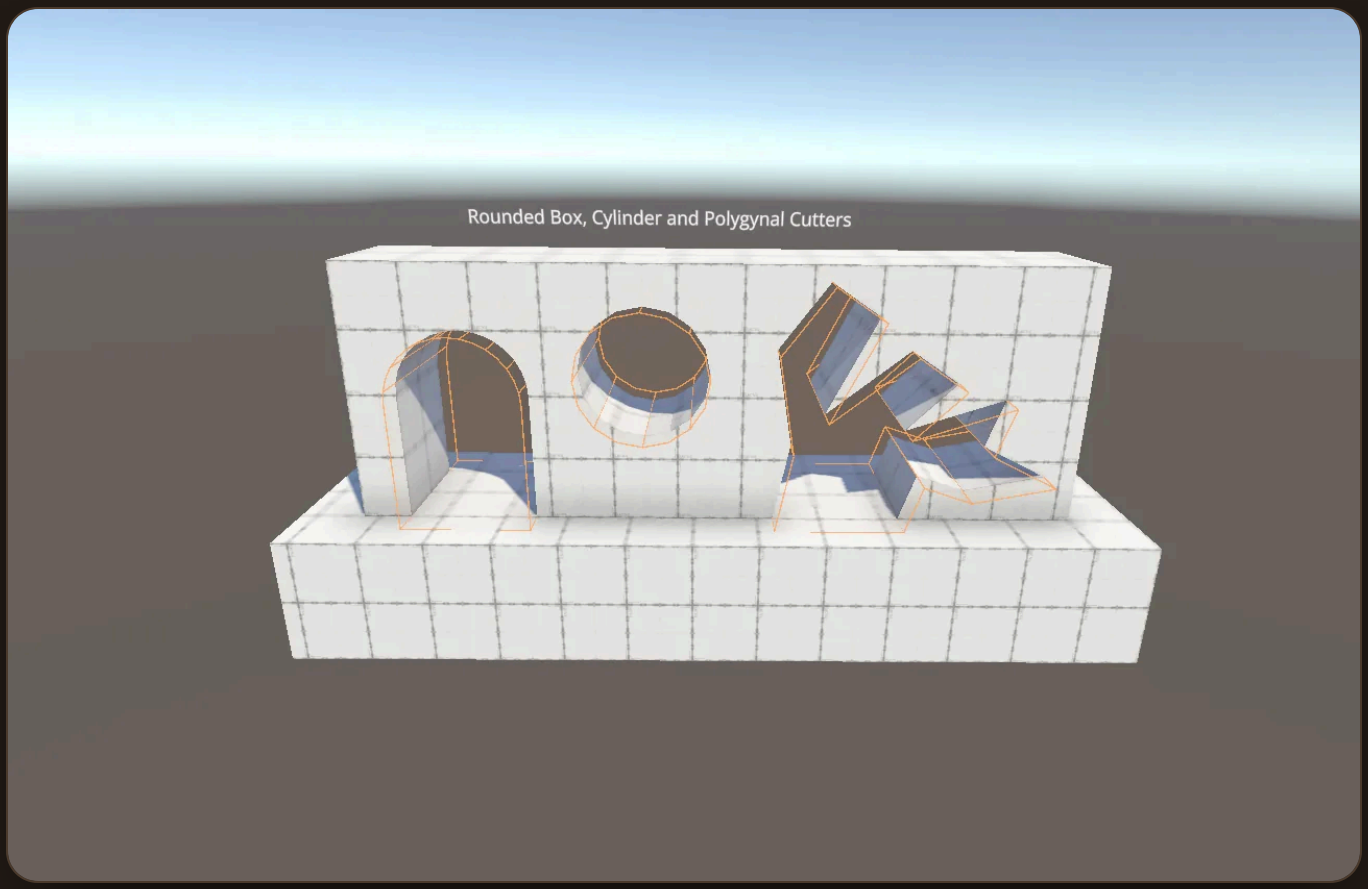
The orange box is the cutter; the hole follows it.

Quick start

1. Create an empty GameObject and place it where you want the hole.
2. Add the component from `Add Component` › `tsitski ProBuilder Tools` › `tsitski ProBuilder Cutter`.
3. Set its **Size** so the box pokes all the way through the wall, then move it into place. The hole appears as soon as the box goes into the mesh.

Shapes

Pick the cutter's shape with **Shape**. Every shape except Path and Mesh is an outline drawn on the cutter's local XY plane and pushed through along its local Z axis, so with no rotation it cuts straight into a wall facing Z.

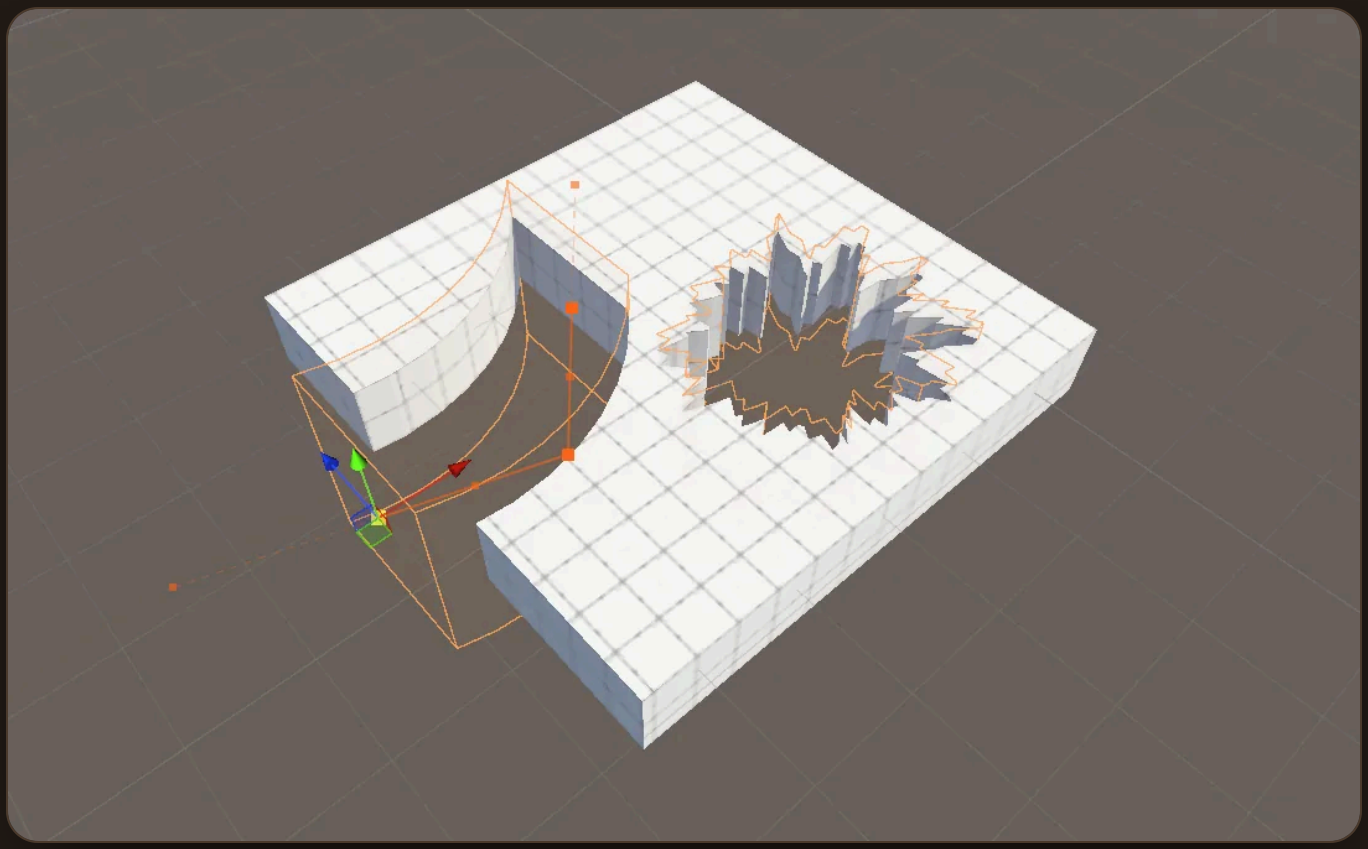


A rounded box, a cylinder and a polygon cutter.

- **Box** — the default. Give any of its four corners a **Corner Radius** to round it. Round just the top two for an arched doorway.
- **Cylinder** — a round hole. A non-square size gives an oval.

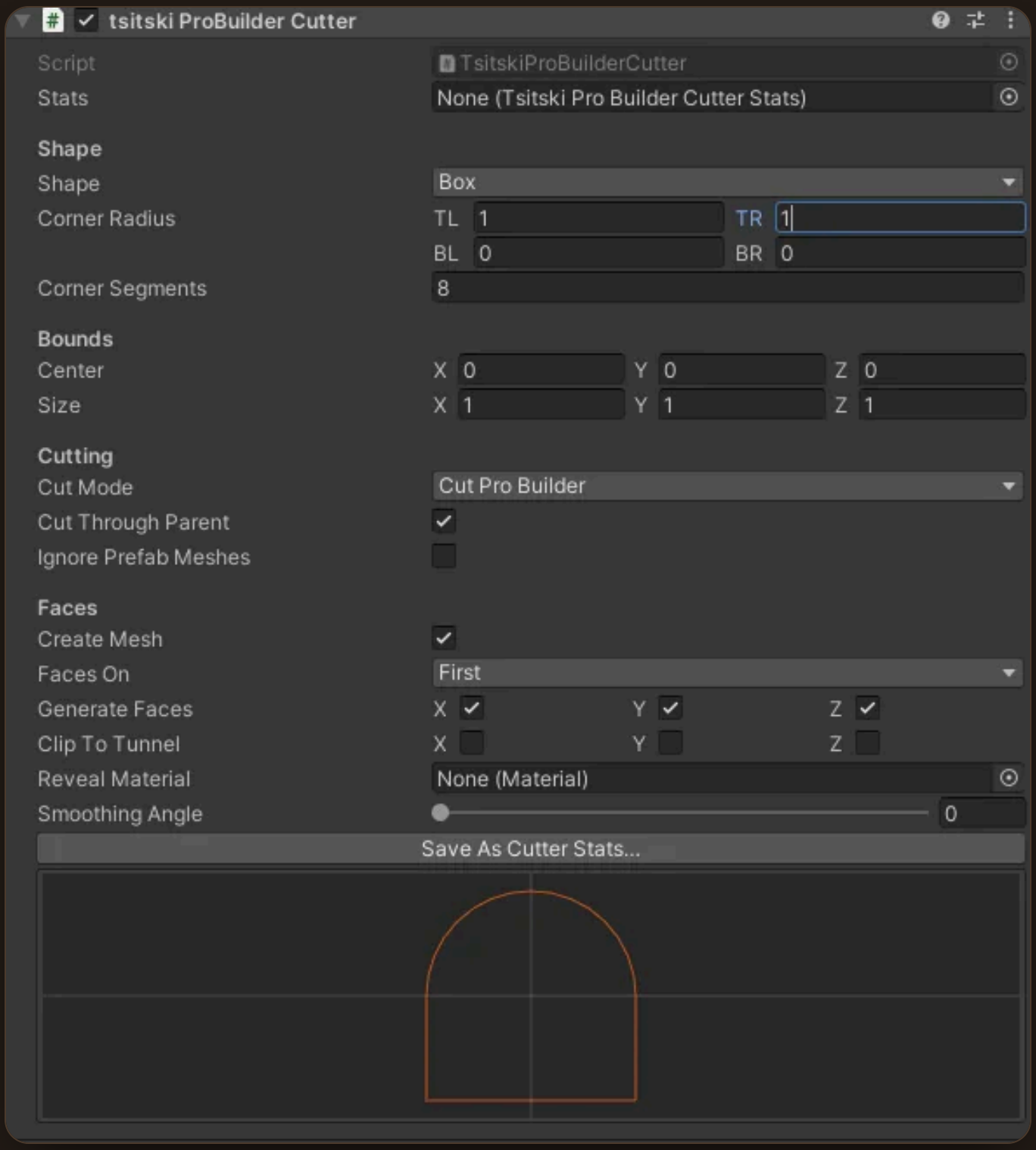
- **Polygon** — any outline you like, including concave ones, edited in the Scene view. Its corners can be rounded too.
- **Path** — a box **Path Width** wide and **Path Height** tall, swept along a path of points in the cutter's local space. Corners can be rounded, and each point can be rotated with the Rotate tool to tilt or roll the box. Great for corridors, trenches and stairwells.
- **Explosion** — a jagged blast hole. Change the seed, spikes and jaggedness, or press **Randomize Explosion**.
- **Mesh** — cuts away an exact mesh in the cutter's local space. Each submesh must be convex; faces shared between submeshes are left out of the cut, so a shape can be built from several convex parts. Spiral stairs generate their stairwell cutter this way.

Polygon and Path points are edited in the Scene view; see Editing points and paths.



A Path cutter and an Explosion cutter.

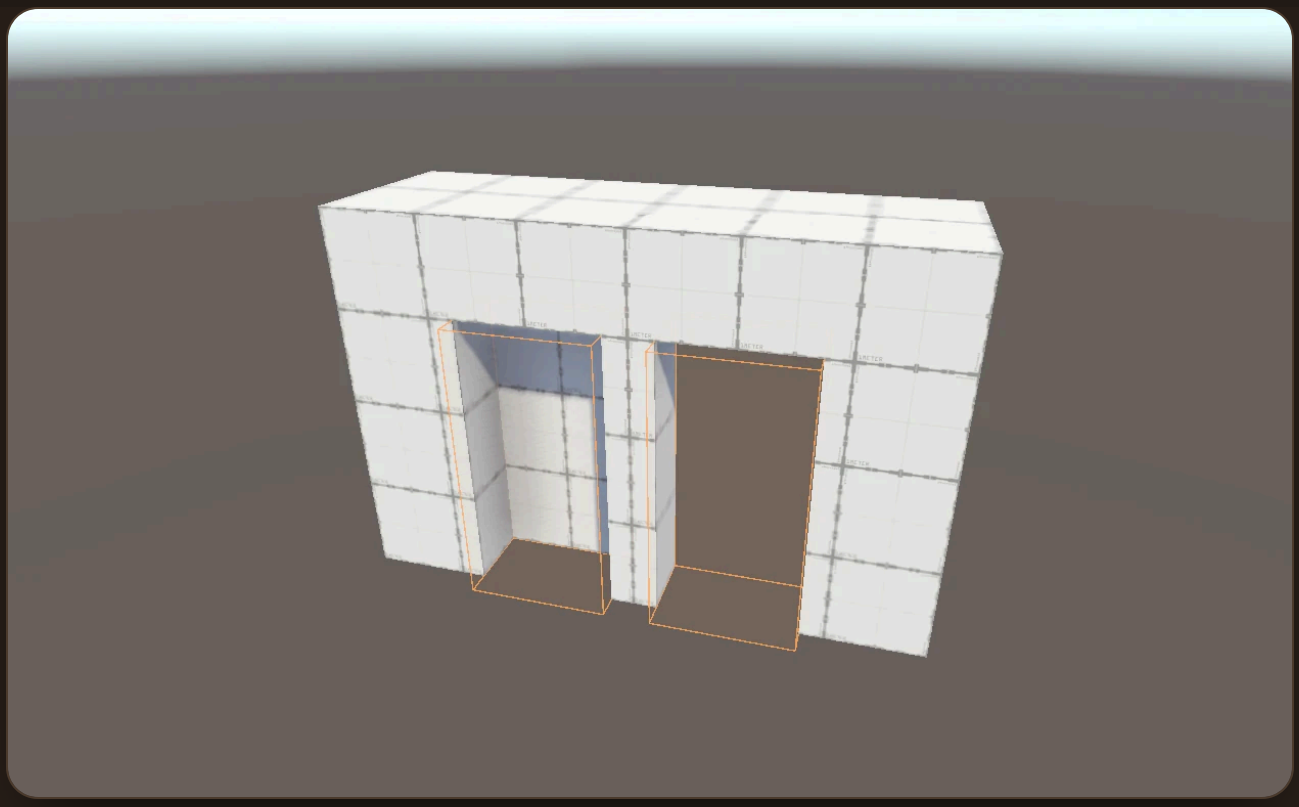
Properties


tsitski ProBuilder Cutter
Script
TsitskiProBuilderCutter
Stats
None (Tsitski Pro Builder Cutter Stats)
Shape
Shape
Box
Corner Radius
TL 1
TR 1
BL 0
BR 0
Corner Segments
8
Bounds
Center
X 0
Y 0
Z 0
Size
X 1
Y 1
Z 1
Cutting
Cut Mode
Cut Pro Builder
Cut Through Parent
Ignore Prefab Meshes
Faces
Create Mesh
Faces On
First
Generate Faces
X
Y
Z
Clip To Tunnel
X
Y
Z
Reveal Material
None (Material)
Smoothing Angle
0
Save As Cutter Stats...

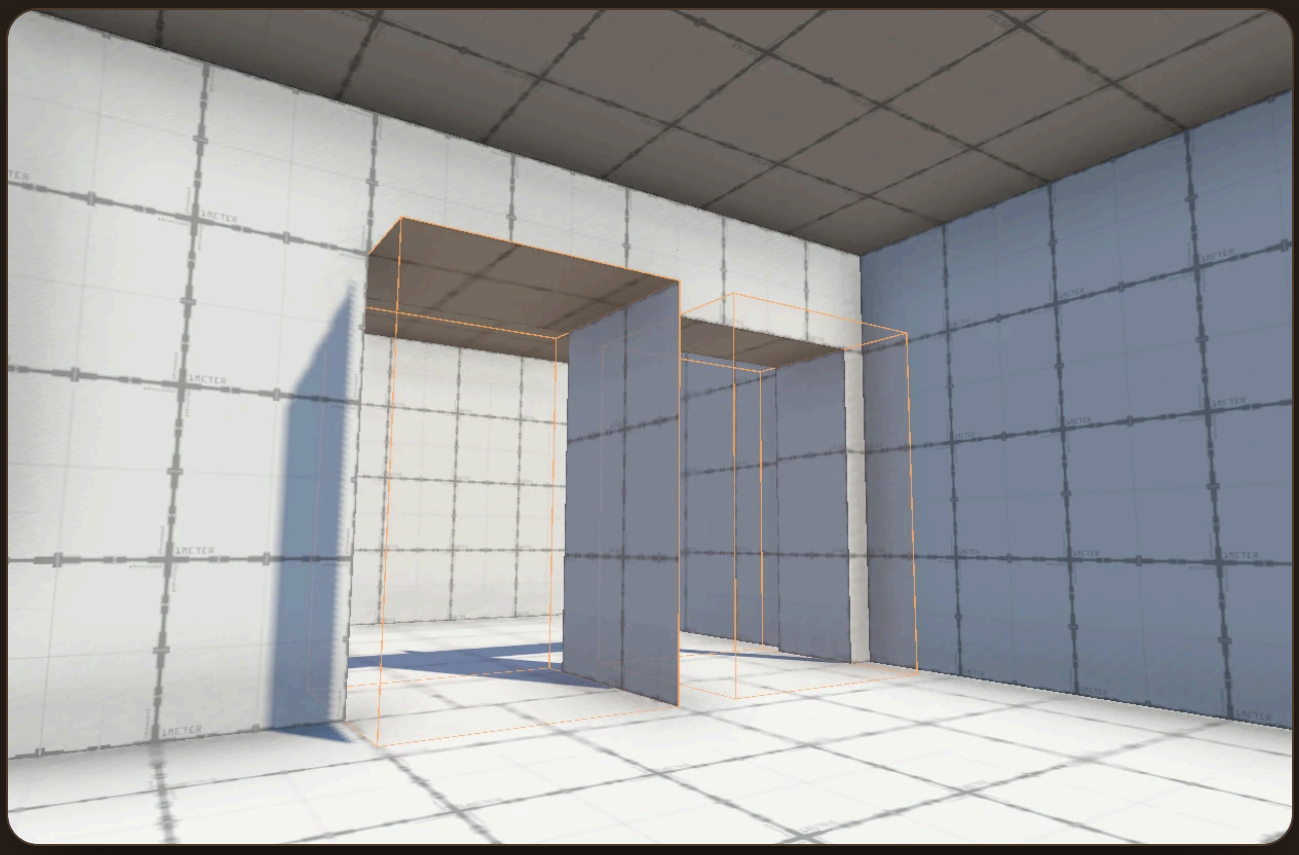
PROPERTY	TYPE	WHAT IT DOES
Stats	Cutter Stats	Shared settings for every cutter using the asset, including the shape, size and points. While it's assigned only this field is shown, and Scene view edits on any of those cutters edit the asset. **Copy Stats Into Cutter** unassigns it and keeps the settings.
Shape	enum	Box, Cylinder, Polygon, Path, Explosion or Mesh. See Shapes above.
Corner Radius	TL / TR / BL / BR	Box only. Rounds each corner, in local units, up to half the box's shorter side.
Corner Segments	int	How many straight pieces each rounded corner is made of (Box, Polygon and Path).
Cylinder Segments	int	Cylinder only. How many sides the cylinder has.
Explosion settings	int, 0–1	Explosion only. Seed, Spikes (12), Minor Spikes (3) and Jaggedness (0.6) shape the blast hole.
Polygon Points	List of Vector2	Polygon only. The outline's points. Easiest to edit in the Scene view.
Path Points	List of Vector3	Path only. The points the box is swept along, centred on them.
Path Rotations	List of Quaternion	Path only. Rotation of the box at each point. Set it with the Rotate tool in the Scene view.
Path Width / Path Height	float	Path only. Size of the swept box.
Shape Mesh	Mesh	Mesh only. The closed mesh to cut away, made of convex submeshes.
Center / Size	Vector3	All shapes except Path and Mesh. The box in local space. A cutter with any size of 0 is ignored.
Smoothing Angle	0–89	Generated faces meeting at less than this angle are shaded smooth. Defaults to 30; 0 keeps every edge hard.

PROPERTY	TYPE	WHAT IT DOES
Create Mesh	bool	Builds the inside walls of the hole. Turn it off to only remove the geometry inside the cutter.
Faces On	enum	Which mesh gets the hole's inside walls when the cutter goes through several meshes. **First** and **Second** put the whole tunnel on the mesh the cutter goes into most or second most deeply, **Shared** has each mesh fill its own part, and **Self** puts them on a ProBuilder mesh under the cutter so every wall just gets a hole.
Cut Mode	enum	Cut ProBuilder meshes, regular meshes (any Mesh Filter with a saved mesh, including generated ones like fences), or both. Regular meshes are never changed; their cut result is generated next to them.
Ignore Prefab Meshes	bool	Don't cut meshes that are part of a prefab instance.
Cut Through Parent	bool	A cutter that's a child of a ProBuilder mesh doesn't cut that mesh, or the other meshes under it, unless this is on. That lets a mesh carry its own cutter without cutting itself.
Generate Faces	X / Y / Z	Which sides of the cutter can become inside walls, per local axis. On a Path cutter, X is the sides, Y the top and bottom and Z the ends.
Clip To Tunnel	X / Y / Z	On that axis, ends the hole's walls at the mesh's own surface instead of at the cutter, so openings finish flush with sloped or uneven walls.
Reveal Material	Material	Material for the new inside walls. Leave it empty to use the mesh's first material.

▾ Generate Faces and Clip To Tunnel, side by side



Generate Faces: each axis toggles a pair of inside walls.

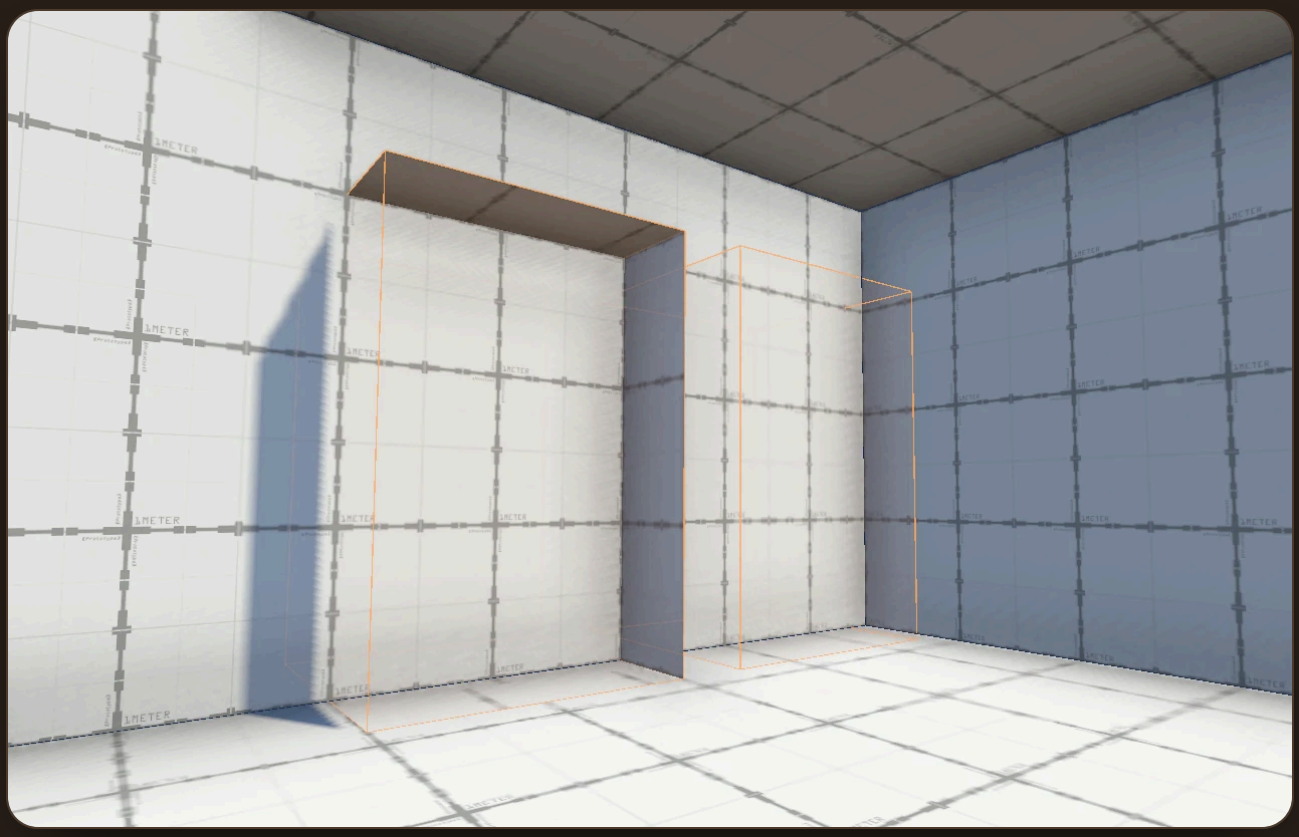


Clip To Tunnel: walls end at the wall's surface instead of the box.

Editing a mesh that has been cut

The cutter never changes your original ProBuilder mesh. The cut result is a separate mesh, created as a child of the original, and shown in its place.

When you select the mesh and switch to a ProBuilder editing mode (vertex, edge or face), the original uncut mesh comes back so you can edit it normally. Switch back to object mode and the cut result returns, rebuilt with your changes.



In ProBuilder's edit modes you edit the original; in object mode you see the cut.

A **ProBuilder Cut Target** component is added to cut meshes automatically to keep track of this, and removed again when nothing cuts them. You never need to add or edit it yourself.

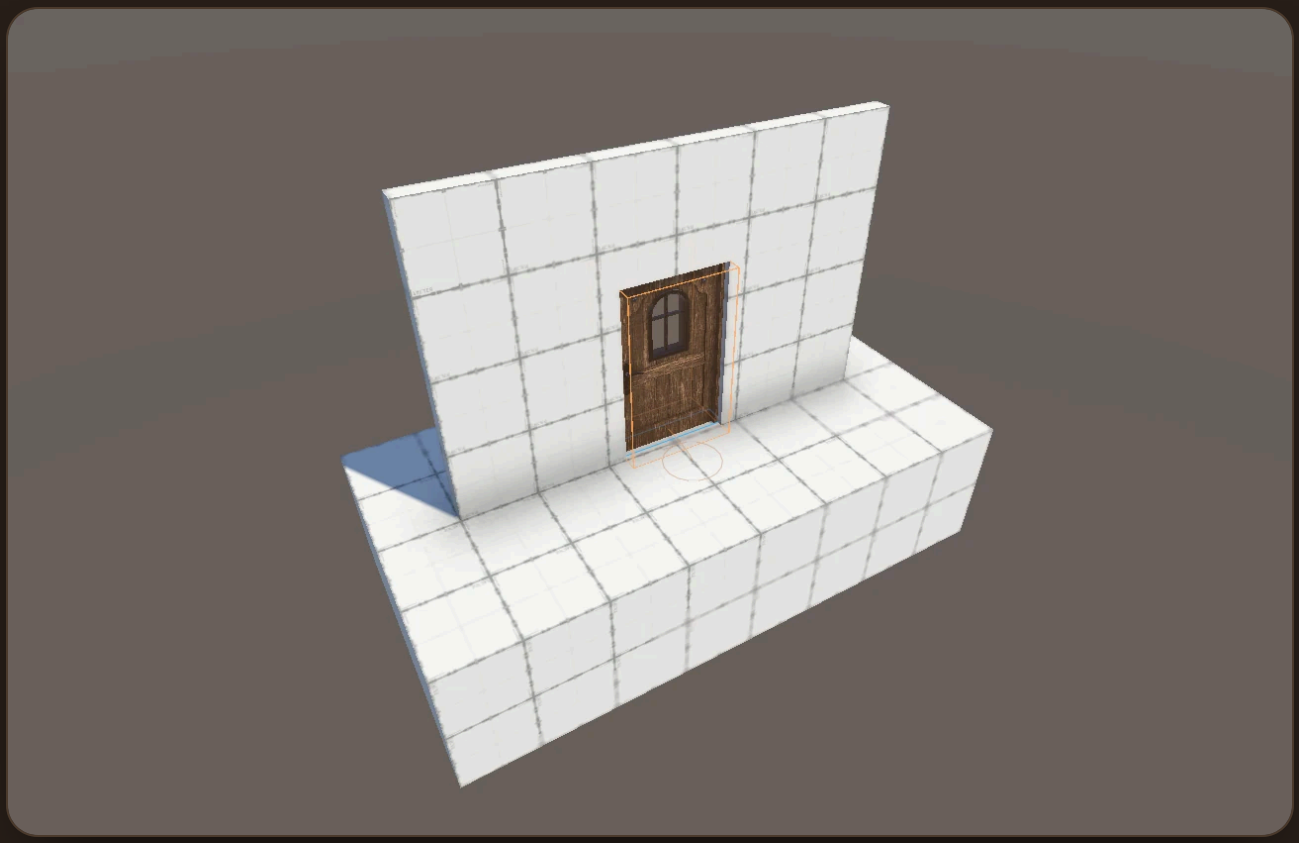
Cutting more than one mesh at once

A cutter cuts every mesh in the same scene that it overlaps (ProBuilder meshes by default; see **Cut Mode**). When it overlaps several, only one of them gets the inside walls,

so you don't end up with doubled walls. By default that's the mesh it goes into most deeply; change it with **Faces On**.

Cutters inside prefabs (doors and windows)

A cutter never cuts meshes that belong to its own prefab or its own children. Put a cutter inside a door or window prefab, drop the prefab against any ProBuilder wall, and it cuts its own opening without cutting the door's own frame.



A window prefab carrying its own cutter.

Colliders, play mode and builds

- The cut result gets a Mesh Collider matching the cut shape, and the original mesh's colliders are switched off while it's cut.
- Cuts are made in the editor and saved with the scene as normal meshes. Nothing runs in play mode or in builds.
- The cut result copies its renderer settings from the original mesh.

Rebuilding every cut

Cuts rebuild by themselves whenever something changes. If a cut ever looks out of date, use `Tools` › `tsitski` › `ProBuilder` › `Generate` › `Rebuild All Cutters`.

Scripting

Everything is in the `Tsitski.ProBuilderTools` namespace. To resize a box cutter from an editor script and have the cut rebuild, use `SetBox`:

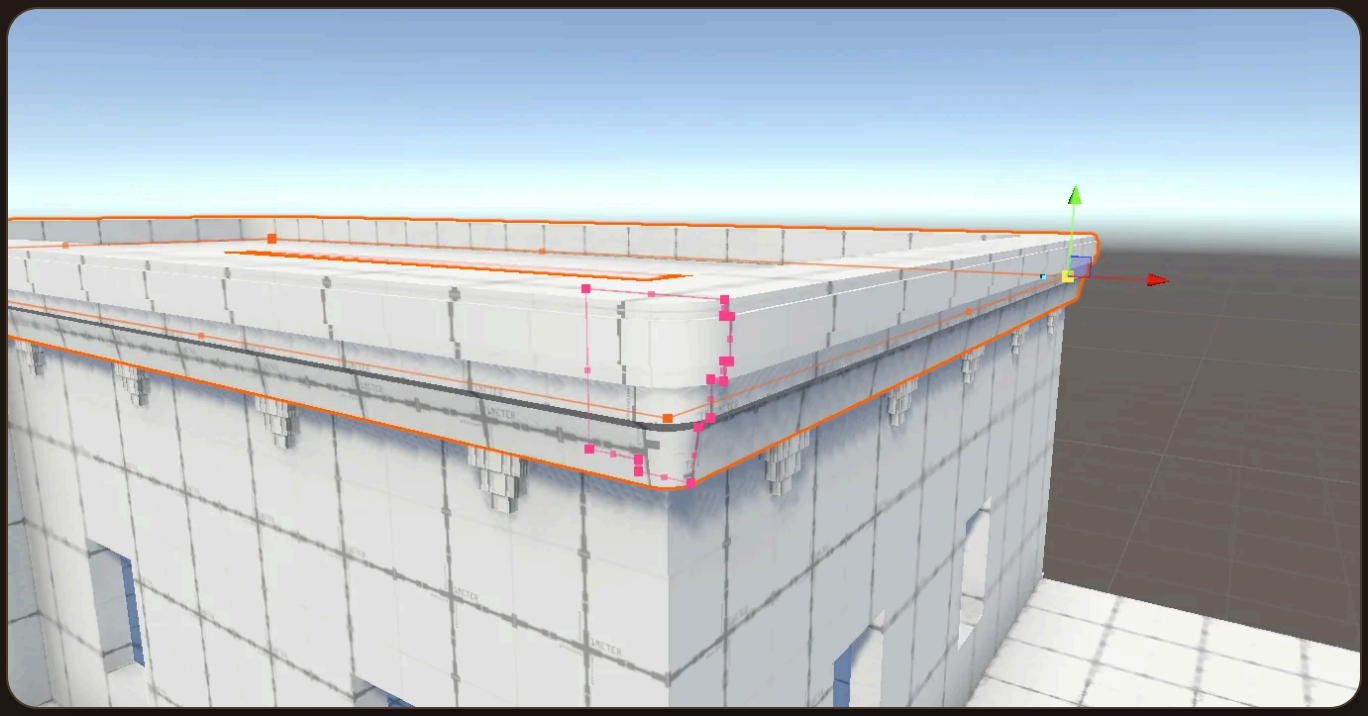
```
using Tsitski.ProBuilderTools;

cutter.SetBox(new Vector3(0f, 1f, 0f), new Vector3(1.2f, 2.1f, 0.6f));
```

After changing other fields from a script, call `cutter.NotifyChanged()`. Editing in the inspector, or moving, rotating or scaling the Transform, rebuilds automatically.

ProBuilder Trim

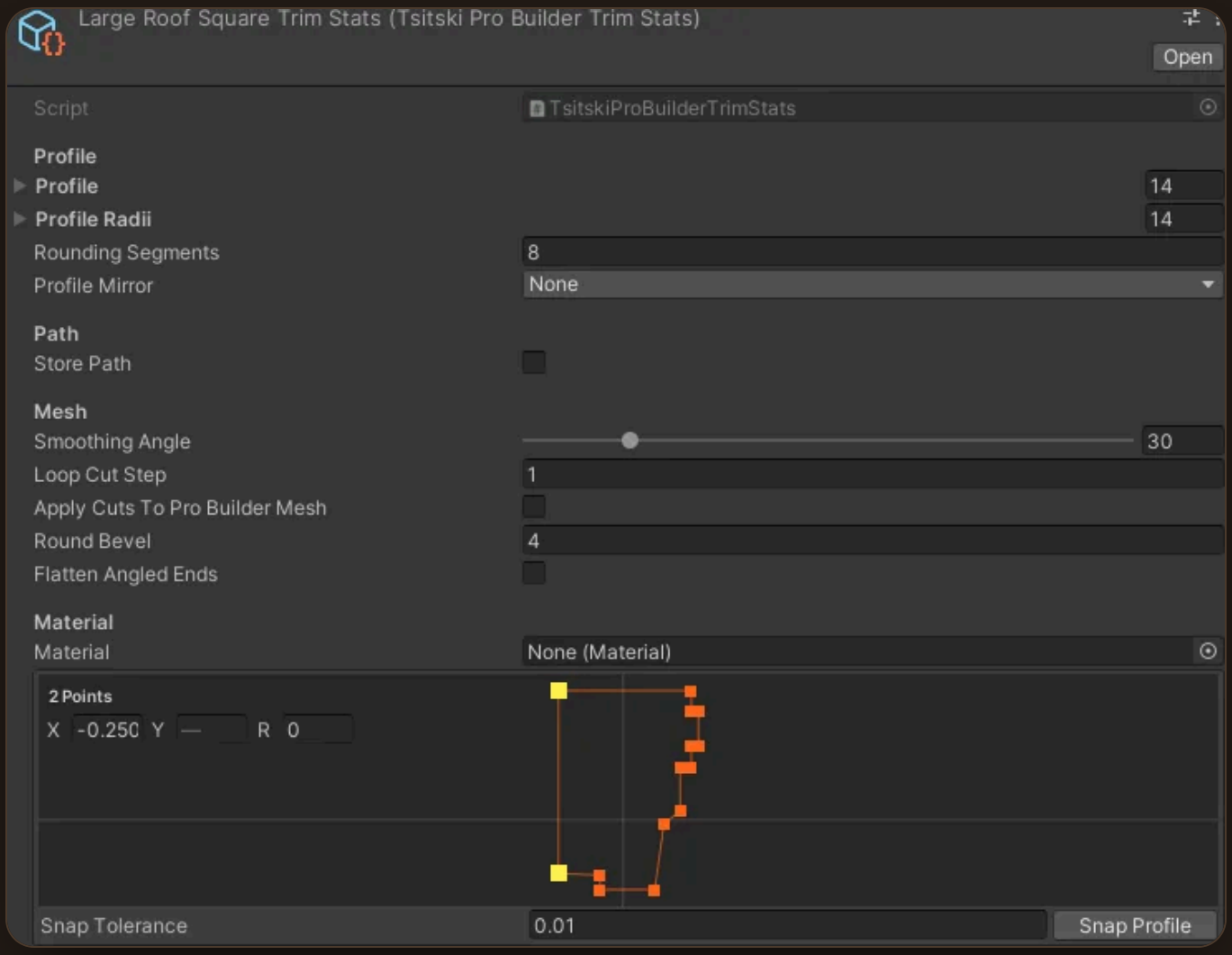
A trim sweeps a 2D profile along a path: skirting boards, crown mouldings, pipes, kerbs, railings. Both the path and the profile are edited in the Scene view, and the mesh rebuilds live.



A trim following a path, with its profile edited at the start.

1. Create an empty GameObject and add `Add Component` › `tsitski ProBuilder Tools` › `tsitski ProBuilder Trim`.
2. Click the dot past the path's end to extend it, and place points where the trim should run.
3. Shape the profile: its points (in pink) are drawn at the start of the path. X points to the right of the path, Y points up, and (0, 0) sits on the path.

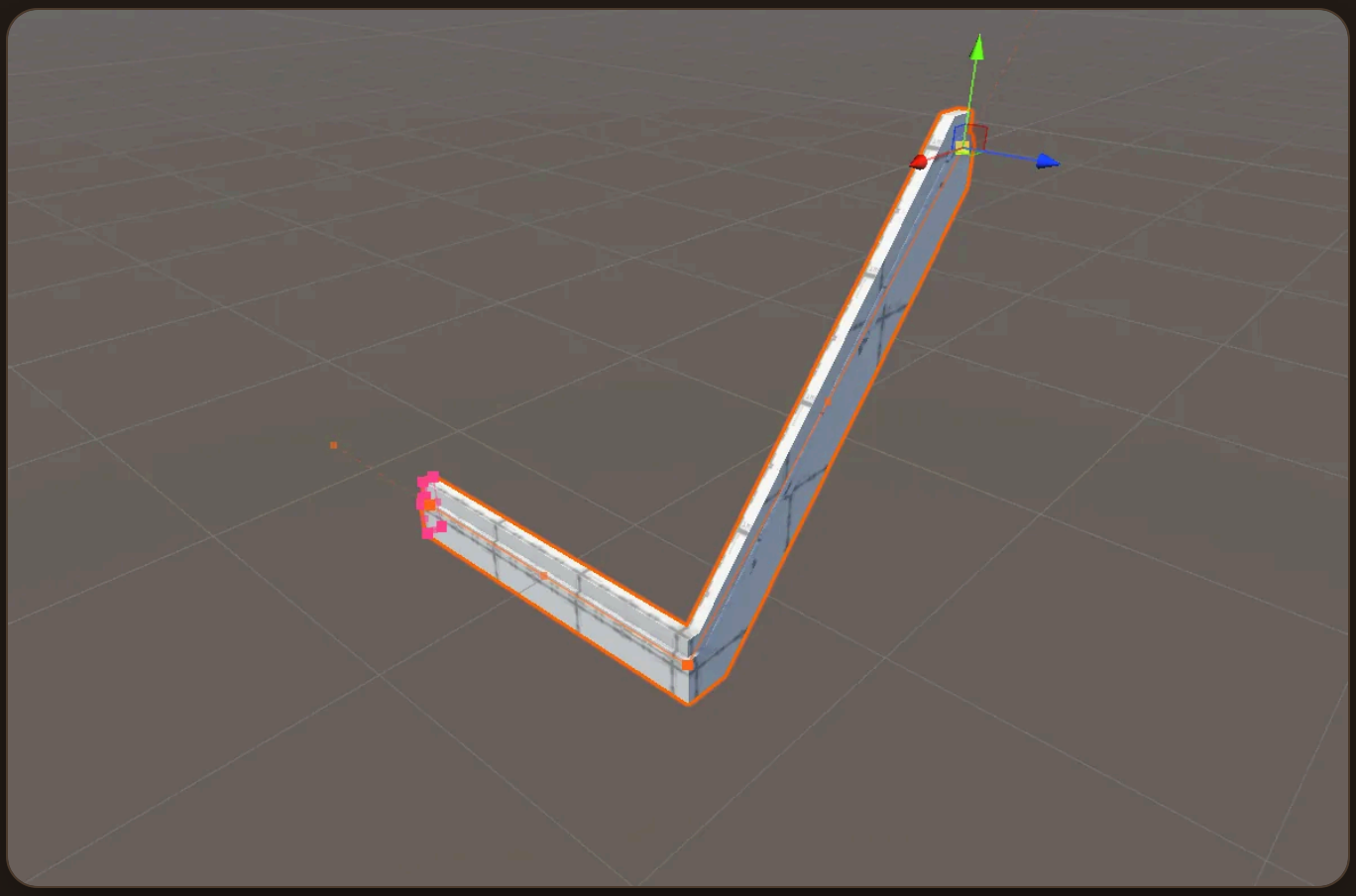
Profile display



The profile display and snap tools.

The inspector draws the finished profile in 2D, and you can edit it right there: click or box-select points, drag them, and type exact positions and corner radii into the fields under the graph. `Esc` clears the selection. Under it, **Snap Profile** lines up profile points that are within **Snap Tolerance** of each other on X or Y, to tidy up hand-placed points.

Sloped paths



The turn is made level, then the slope continues.

Where a path both turns and climbs at a sharp corner, the turn is made level at the corner's height with a clean mitre, and the slope starts just after it, so the profile never twists. Rounded corners stay smooth curves. **Flatten Angled Ends** does the same at the start and end of a sloped path so the end caps stand upright.

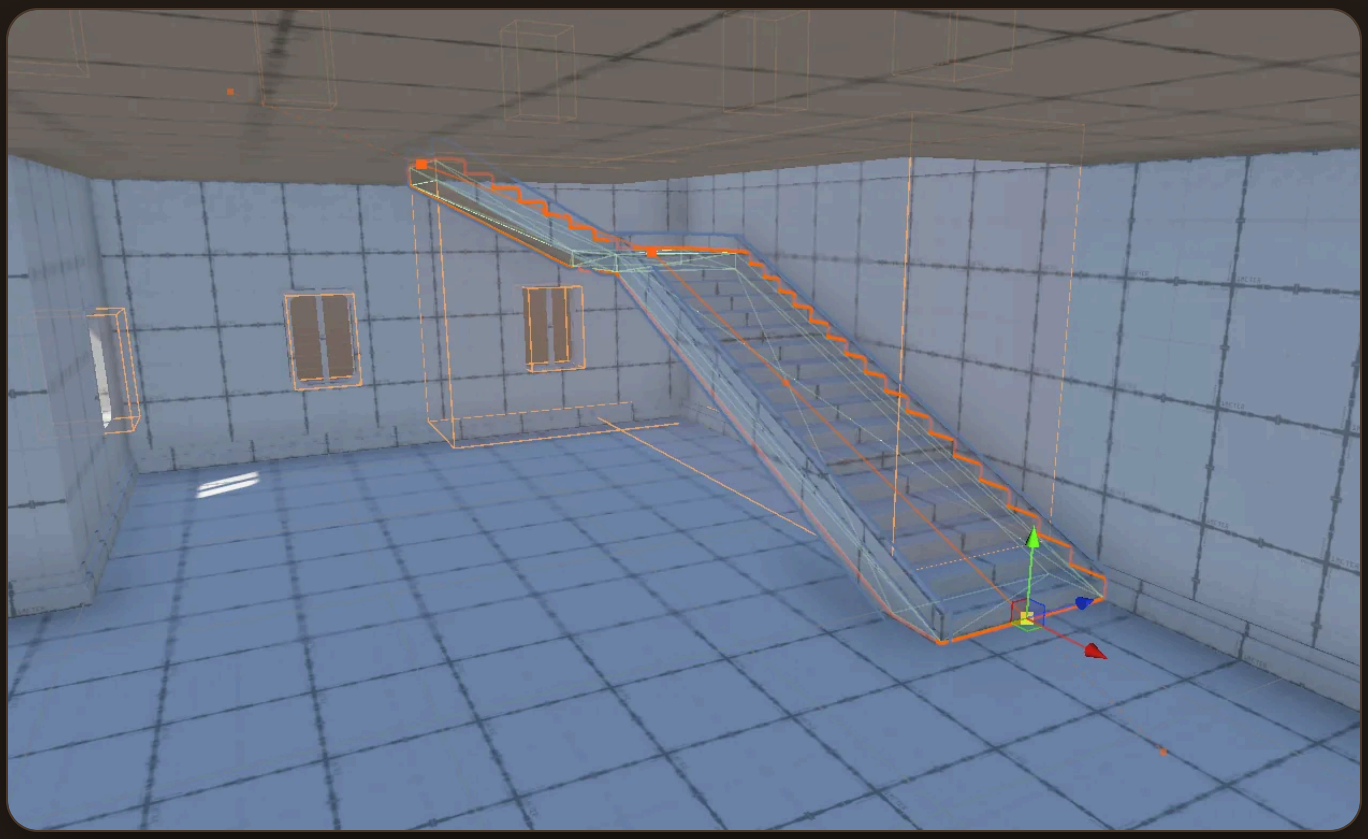
PROPERTY	TYPE	WHAT IT DOES
Stats	Trim Stats	Shared profile and settings. With **Store Path** on, the asset holds the path too.
Material	Material	Locked onto every face and reapplied on each rebuild. Empty leaves the faces free. See Materials.
Profile / Profile Radii	List of Vector2 / float	The cross-section and the rounding of each of its corners.
Path / Path Radii	List of Vector3 / float	The path in local space, and the rounding of each corner.
Closed	bool	Joins the last point back to the first.
Cap Ends	bool	Closes the ends of an open trim.
Smoothing Angle	0–180	Faces meeting at less than this angle are shaded smooth.
Loop Cut Step	float	Distance between extra loop cuts along each segment. Below 0.1 turns them off.
Apply Cuts To ProBuilder Mesh	bool	Makes every loop cut its own ProBuilder face. Off keeps the ProBuilder faces whole and only splits the triangles of the built mesh.
Round Bevel	int	Segments used to round every path corner. 0 keeps sharp mitred corners.
Rounding Segments	int	How smooth rounded path and profile corners are.
Profile Mirror	enum	None, Across Y, Across X or Both. Draw half the profile and have it mirrored to complete the shape.
Flatten Angled Ends	bool	Runs the trim level for a short way at sloped ends so the caps are upright.

TIP

To cut a trim in two, select some of its path points and run `Tools` › `tsitski` › `Core` › `Split Path`. See Split Path.

ProBuilder Stairs

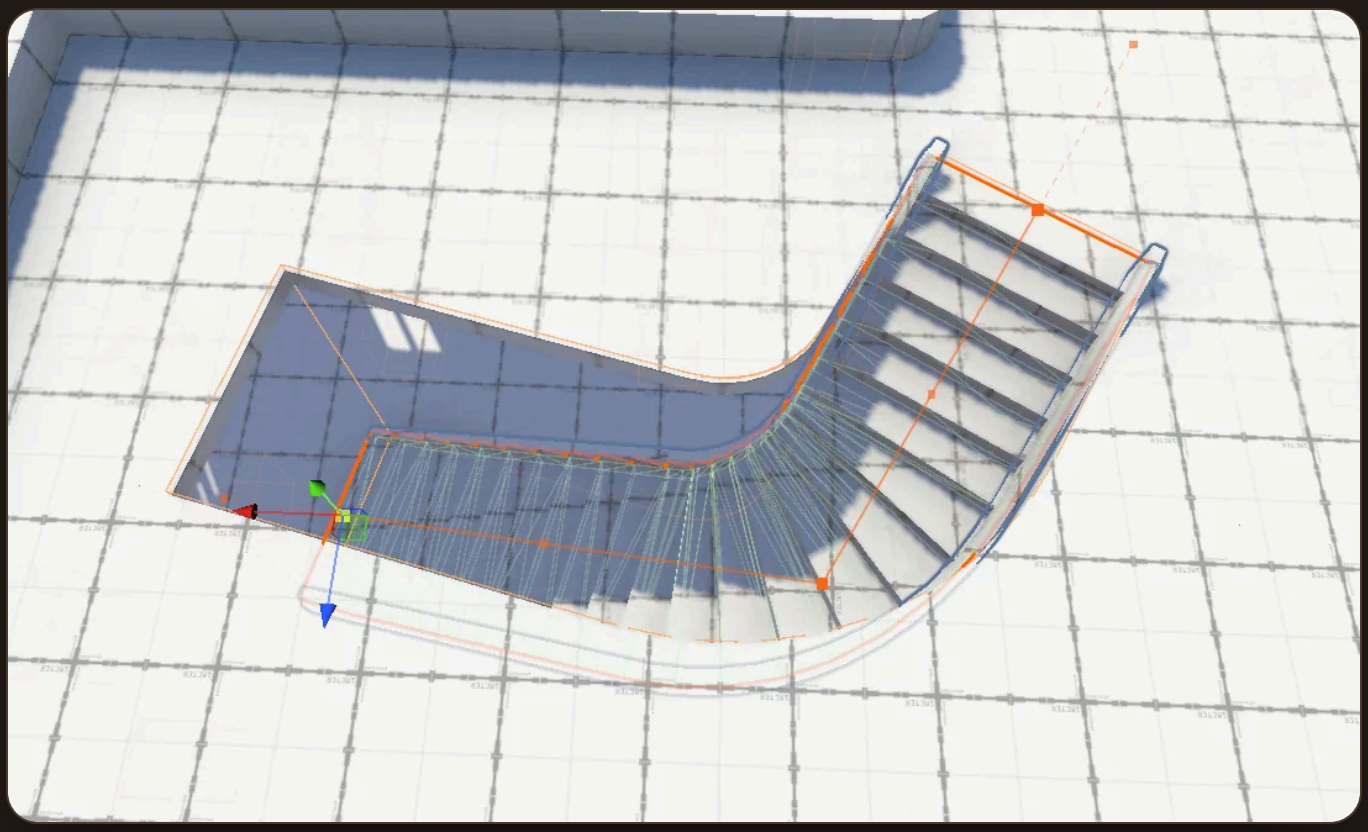
Stairs have two modes. In **Path** mode (the default) they follow a path: each pair of points is a flight, the first point is the front edge of the first step, the last point is the back edge of the last step, and every point in between gets a square landing at its height. In **Spiral** mode they wind around the object's pivot (see Spiral stairs below).



Stairs with a landing on the middle point.

1. Add `Add Component` › `tsitski ProBuilder Tools` › `tsitski ProBuilder Stairs` to an empty GameObject.
2. Move the path's points to set where the stairs start and end, and add points for landings.
3. Round a corner point to swap its landing for winding steps.

Landings and winding corners



A rounded corner turns the landing into winding steps.

- A 90° turn gets a full square landing, so L-shaped stairs just need three points.
- Rounding a point replaces its landing with winding steps. The radius is the inner edge's radius, so a tiny radius pivots the steps around a point and a bigger one leaves an open well.
- Flights joined by rounded points climb evenly from one landing to the next, so a rounded point's own height is ignored.
- The step count is rounded so every step fits exactly; the inspector shows the total.

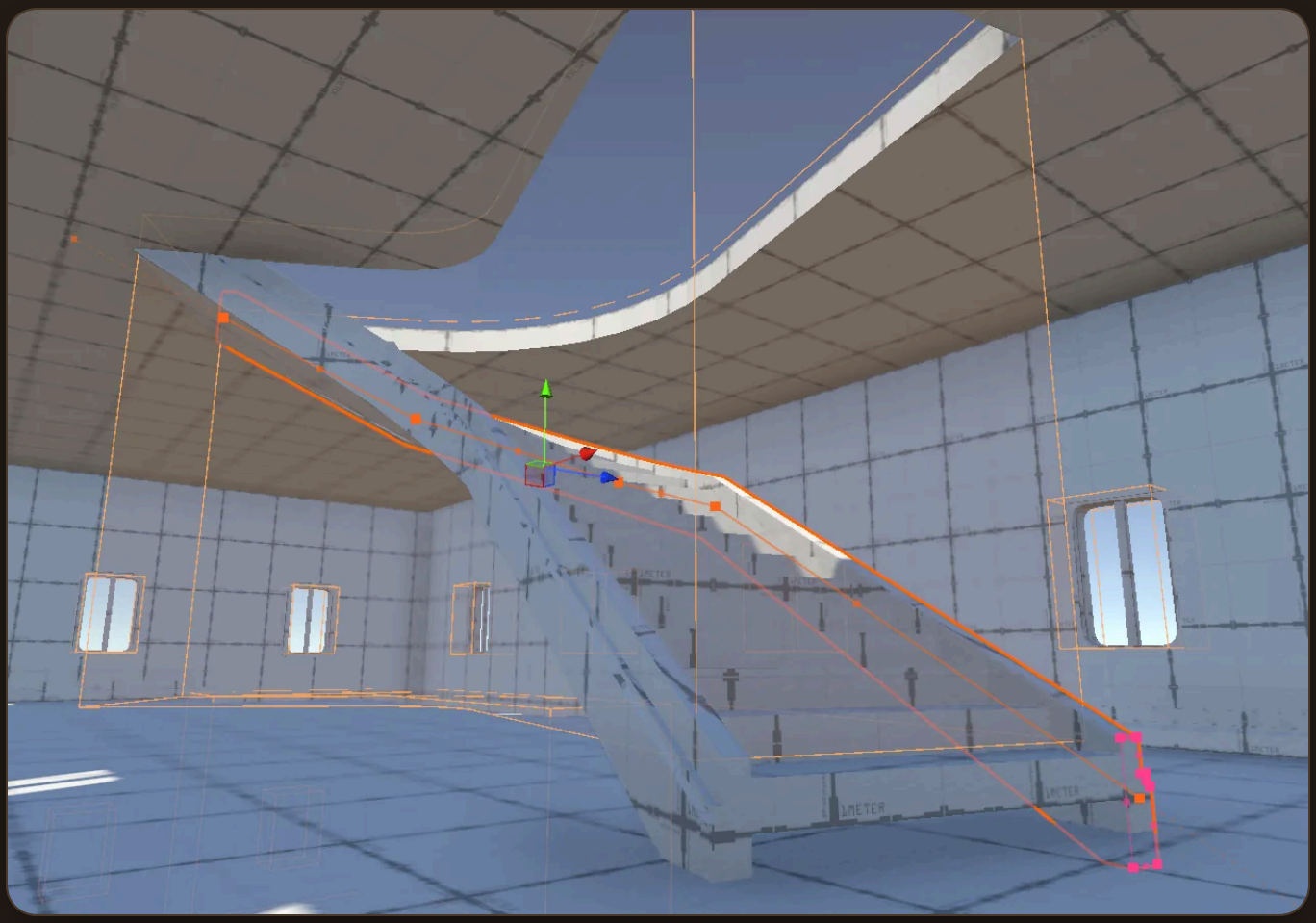
Spiral stairs

Set **Mode** to **Spiral** and the path is replaced by a spiral around the object's pivot, set by **Spiral Height**, **Spiral Radius** (from the centre to the middle of the steps), **Width** and **Spiral Angle**: 90 for a quarter turn, 360 for a full turn, or more for several turns. **Spiral Clockwise** picks the direction. A radius of half the width or less closes the centre.

- Drag the arrow up the centre for the height, and the handle along the bottom step for the radius.

- Drag the side handles across the bottom step for the width.
- Drag the handle on the top step around the spiral to change the angle; it keeps counting past 360°.
- Solid suits turns under 360°. For a full turn or more, turn Solid off so the steps don't fill down over the turn below.

Stairwell cutter, trims and collider



Generate Cutter opens a stairwell; side trims follow both edges.

- **Generate Cutter** adds a cutter following the stairs, so the floor above gets a stairwell cut through it. It only covers the part of the stairs within **Cutter Headroom** of the top, and never cuts the stairs themselves or the floor they start on. Path stairs get a flat Path cutter; spiral stairs get a Mesh cutter built from the steps themselves, so the opening matches the top step exactly.

- **Left Trim** and **Right Trim** take Trim Stats and generate a trim along each edge of the stairs and landings, curving round winding corners and following the exact spiral on spiral stairs. The profile's X points away from the stairs on both sides, so one asset makes a mirrored pair.
- **Smooth Collider** gives the Mesh Collider a ramp instead of steps, so characters walk up smoothly. The visible stairs keep their steps.

The treads and underside of winding and spiral stairs are UV mapped along the stairs, so textures run smoothly round the curve instead of changing direction on every face.

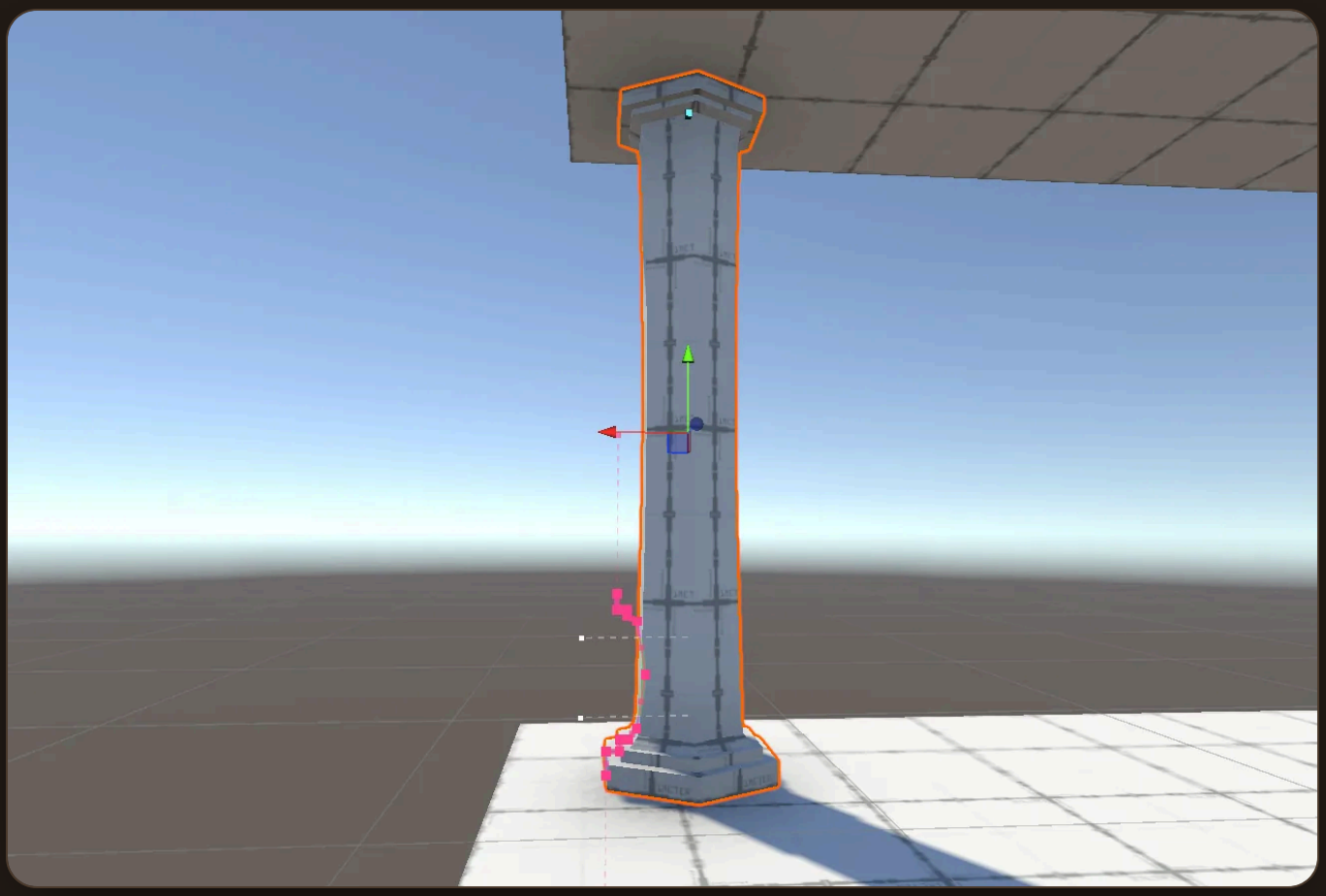
PROPERTY	TYPE	WHAT IT DOES
Stats	Stairs Stats	Shared width, step height, solid, thickness, cutter, trim and collider settings.
Top Step / Forward Step / Side Material	Material	Locked onto the treads and landing tops, onto the risers, and onto the sides, underside, back and landing edges. See Materials.
Mode	enum	Path follows the path points; Spiral winds around the pivot.
Spiral Height	float	Spiral only. Total height the spiral climbs.
Spiral Radius	float	Spiral only. Distance from the centre to the middle of the steps.
Spiral Angle	10–1440	Spiral only. How far the spiral turns, in degrees.
Spiral Clockwise	bool	Spiral only. Turn clockwise (seen from above) while climbing.
Path / Path Radii	List	Path only. The flights and corner rounding.
Width	float	Width of the stairs and landings.
Step Height	float	Target riser height.
Solid	bool	Fills down to the lowest path point. Off gives a sloped underside instead.
Thickness	float	Underside and landing thickness when Solid is off.
Generate Cutter	bool	Adds the stairwell cutter.
Cutter Headroom	float	How far below the top of the stairs the stairwell cutter reaches.
Left Trim / Right Trim	Trim Stats	Generates side trims.
Smooth Collider	bool	Ramp collider instead of steps.

NOTE

The cutter and trims are generated children of the stairs and rebuilt with them, so don't edit them by hand.

ProBuilder Pillar

A pillar is built from three profiles: a **bottom** (base) and **top** (capital) with their own heights, and a **center** (shaft) that repeats, or stretches, to fill the rest of the pillar's height. It's either spun into a round or square pillar, or swept around a wall corner.



A 12-sided pillar and a square pillar from the same profile.

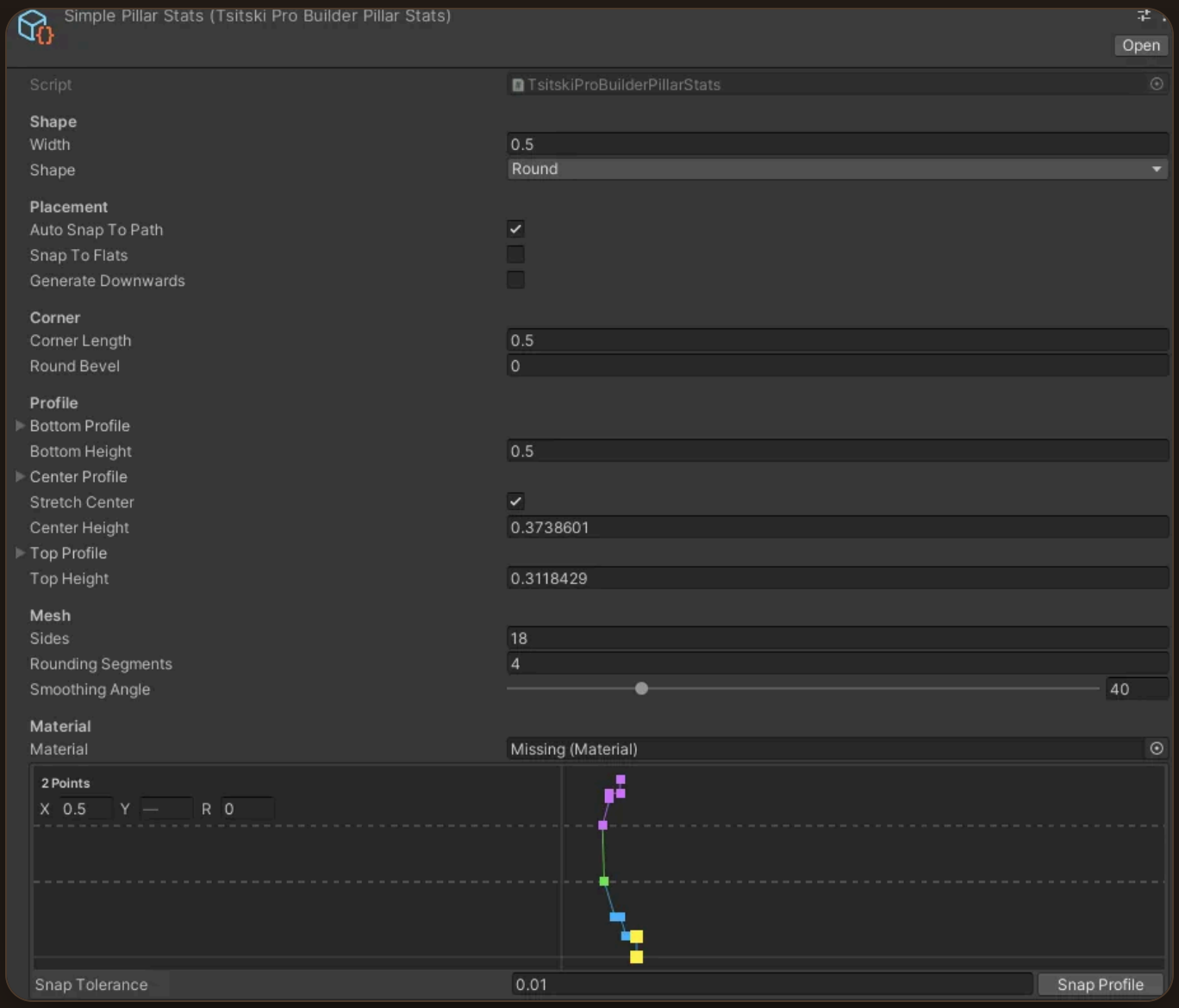
1. Add `Add Component` › `tsitski ProBuilder Tools` › `tsitski ProBuilder Pillar` to an empty GameObject.
2. Edit each profile's points on the pillar's +X side, coloured blue (bottom), green (center) and purple (top). X is the distance from the axis, Y runs 0 to 1 across that part's height.
3. Drag the cube handles on the dotted split lines to change the bottom and top heights, and the handle on top of the pillar to set its total height.

Round and corner pillars

- **Round** spins the profiles around the pillar's axis. **Sides** sets how many sides it has; 4 makes a square pillar.
- **Corner** sweeps the profiles along two legs, from **Corner Start** through the pillar's origin to **Corner End**, so the pillar wraps the corner of a wall. Drag the leg ends in the Scene view. **Round Bevel** rounds the outer side of the bend.

Snapping to paths

With **Auto Snap To Path** on, moving a pillar snaps it to the nearest corner of a floor plan, trim, array or stairs path within **Snap Distance**, and a corner pillar turns its legs to follow the path. **Snap To Flats** also snaps onto straight parts of a path. The **Snap To Path** button snaps it once.



Bottom, center and top profiles, edited in the inspector graph.

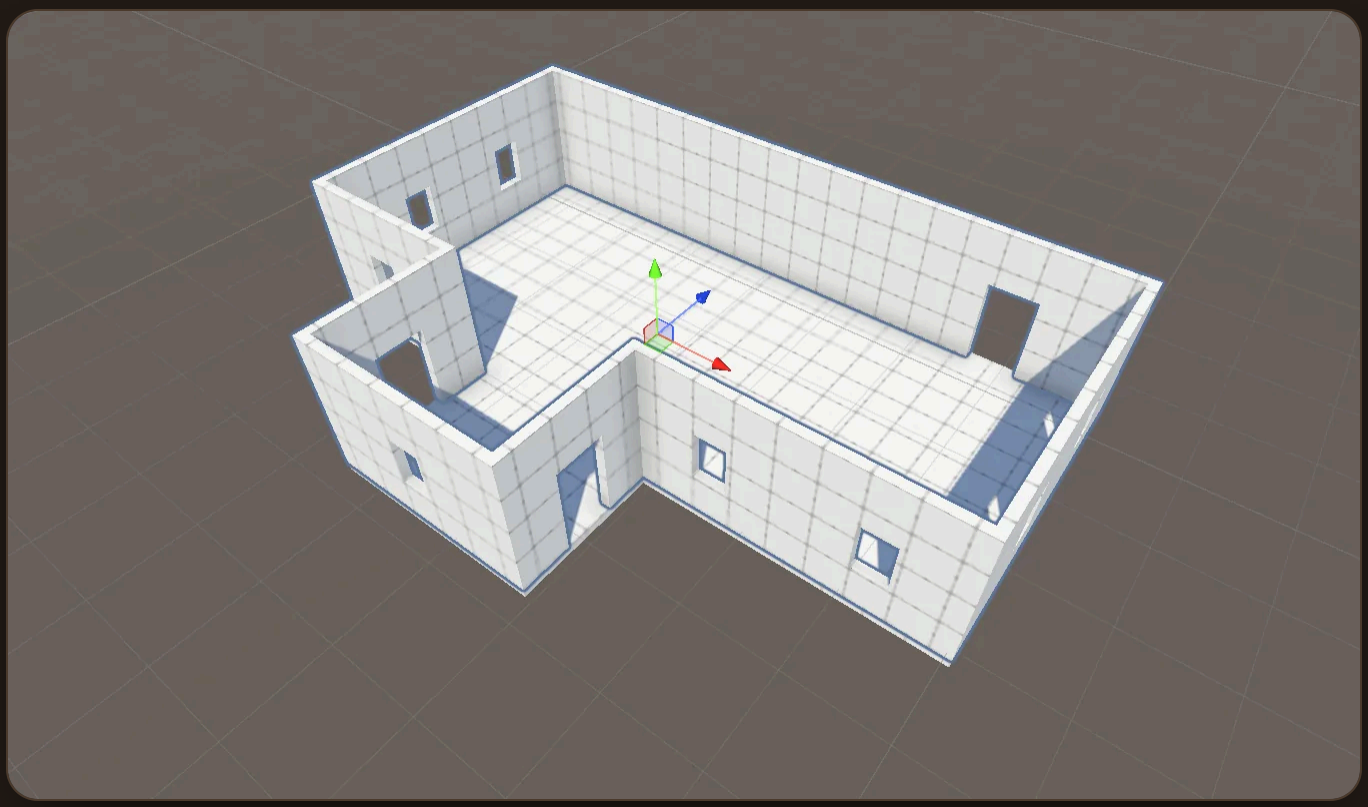
The inspector graph shows all three profiles stacked at their real heights, and can be edited directly like the trim profile display. Repeated center sections are drawn faded.

PROPERTY	TYPE	WHAT IT DOES
Stats	Pillar Stats	Shared profiles, heights, sides and smoothing.
Material	Material	Locked onto every face. See Materials.
Height	float	Total height of the pillar.
Width	float	Distance of the furthest profile point from the axis. Changing it scales every profile point in or out.
Shape	enum	Round or Corner.
Auto Snap To Path / Snap To Flats	bool	Snap to path corners, and to straight parts of paths.
Generate Downwards	bool	Hang the pillar down from its origin instead of standing up from it.
Corner Side	enum	Corner only. Outside or Inside: which side of the legs the pillar sits on.
Corner Length	float	Corner only. Length of both legs.
Round Bevel	int	Corner only. Segments rounding the outer side of the bend. 0 keeps a sharp mitre.
Corner Start / Corner End	Vector3	Corner only. Directions of the two legs from the origin.
Snap Distance	float	How close a path corner has to be to snap to it.
Bottom / Center / Top Profile	profile	The three outlines, each with its own points and corner radii.
Bottom Height / Top Height	float	Heights of the base and capital in metres. Squished if the pillar is shorter than both together.
Center Height	float	Height of one repeat of the center profile. The repeat count is rounded so whole repeats fit.

PROPERTY	TYPE	WHAT IT DOES
Stretch Center	bool	Stretch the center profile once instead of repeating it.
Sides	int	Round only. Number of sides. 4 makes a square pillar with its faces along X and Z.
Rounding Segments	int	How smooth rounded profile corners are.
Smoothing Angle	0–180	Sides and profile bends under this angle are shaded smooth, so square pillars stay crisp.

ProBuilder Floor Plan

Generates walls along a path, or around the outline of a floor, and keeps them up to date as you edit. Layers add trims and arrays along the wall faces, and a Roof Stats asset puts a roof on top.



Walls following the outline of a floor.

1. Add `Add Component` › `tsitski ProBuilder Tools` › `tsitski ProBuilder Floor Plan` to an empty GameObject and edit its path in the Scene view, or select a ProBuilder floor and run `Tools` › `tsitski` › `ProBuilder` › `Generate` › `Walls From Floor Plan` to have the walls follow its outline.
2. Set the wall height, thickness and alignment.
3. Add layers for skirting, cornices or corner pillars, and a Roof Stats asset for a roof.

Path or floor

With **Floor** empty, the walls follow the floor plan's own path, edited like any other path (see Editing points and paths). Open paths work too, with end caps on the walls. With a **Floor** assigned, the path is generated from the outline of the floor's top faces and drawn read-only, and the walls rebuild whenever the floor changes.

Layers

Each layer generates a trim (from **Trim Stats**) and/or an array (from **Array Stats**) along the wall faces, as child objects. **Wall Side** picks the outside faces, the inside faces or both, and **Height** sets how far up the wall it runs, or down from the top with **Height From Top**. Array Stats with corner Pillar Stats put a pillar in every corner.

Roof

Assign a **Roof Stats** asset to generate a roof over every closed wall loop, sitting on top of the walls. The roof's first path sets the slope from its Width and Height. See ProBuilder Roof.

Removed layers

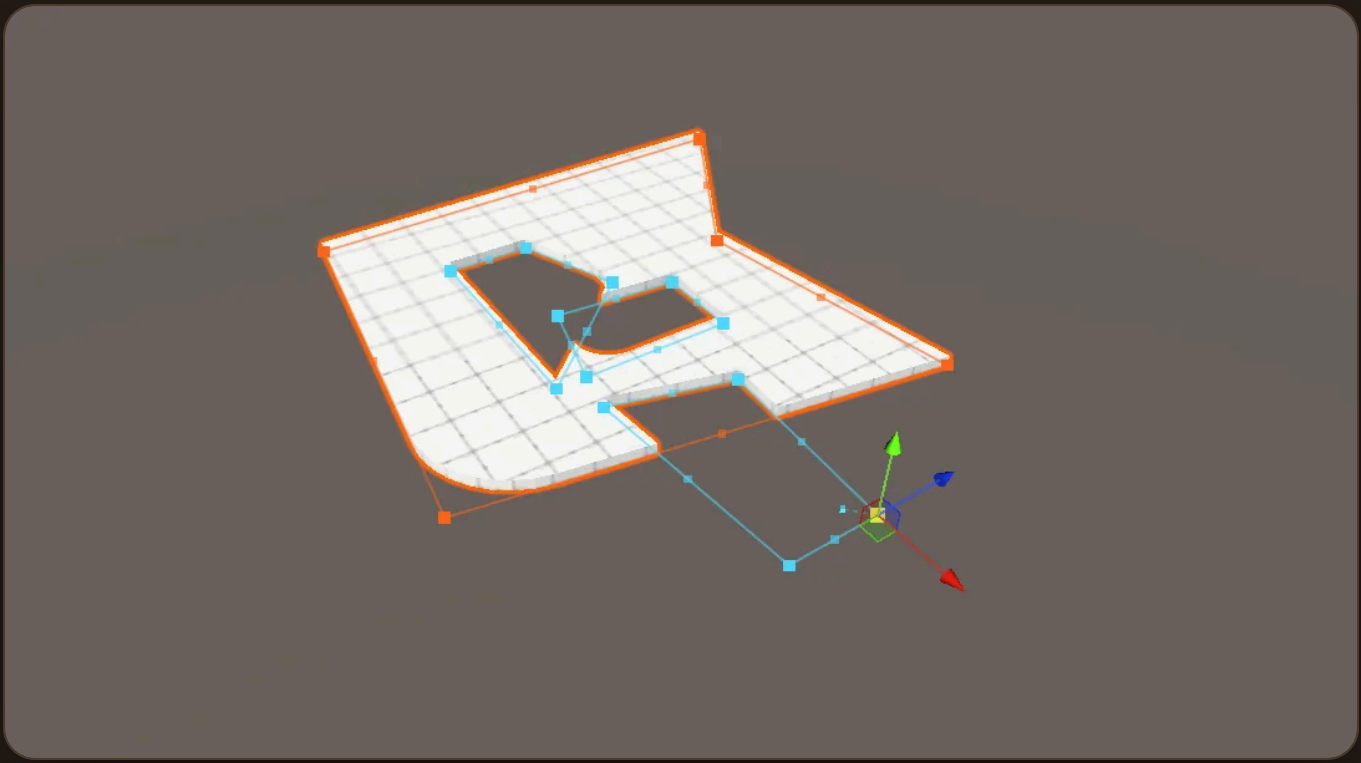
Delete a generated trim, array or roof and it stays deleted when the floor plan rebuilds, without changing the stats. Deleted parts are listed under **Removed Layers** in the inspector, with **Restore** and **Restore All** buttons.

PROPERTY	TYPE	WHAT IT DOES
Stats	Floor Plan Stats	Shared alignment, corner, loop cut, layer and roof settings, and the height and thickness while Store Height and Store Thickness are on.
Inside / Outside Material	Material	Locked onto the inside and outside faces of the walls. See Materials.
Floor	ProBuilder Mesh	Optional. When set, the path follows this floor's outline.
Path / Path Radii	List	The path the walls follow, and the rounding of each corner.
Closed	bool	Joins the last point back to the first.
Height	float	Wall height.
Thickness	float	Wall thickness.
Alignment	enum	Inside, Center or Outside: which side of the path the walls sit on.
Walls Around Holes	bool	Also builds walls around holes in the floor.
Square Corners	bool	A separate block at each corner instead of a mitred joint.
Loop Cut Step / Vertical Loop Cut Step	float	Distance between extra loop cuts along and up the walls. Below 0.1 turns them off.
Apply Cuts To ProBuilder Mesh	bool	Make every loop cut its own ProBuilder face instead of only splitting the built mesh.
Rounding Segments	int	How smooth rounded corners are.
Roof Stats	Roof Stats	Generates a roof on top of every closed wall loop. Empty builds no roof.

PROPERTY	TYPE	WHAT IT DOES
Layers	List	Trims and arrays along the wall faces: Trim Stats, Array Stats, Wall Side, Height and Height From Top.

ProBuilder Polygon

One or more outlines drawn in the Scene view and extruded into a slab, with any number of holes. Use it for floors, ceilings, platforms and flat roofs.



A polygon outline edited in the Scene view.

1. Add `Add Component` › `tsitski ProBuilder Tools` › `tsitski ProBuilder Polygon` to an empty GameObject, or select existing ProBuilder meshes and run `Tools` › `tsitski` › `ProBuilder` › `Generate` › `Convert To Polygon` to turn their top outline (and its holes) into a polygon.
2. Edit the outline's points in the Scene view (see Editing points and paths).
3. Click **Add Hole** to add a 2 by 2 hole in the middle of the outline, then edit its points the same way.

4. Click **Add Path** to add another outline next to the shape, or copy a whole outline and paste it.
5. Drag the arrow above the shape to set the thickness. Drag it below the outline to extrude downwards.

Outlines that overlap or touch are merged into one shape, and outlines that don't stay separate pieces of the same mesh. Holes are cut from the merged shape.

Holes can go anywhere. Holes that overlap or touch are merged into one opening, a hole that crosses the outline cuts a notch out of it, and a hole that cuts all the way across splits the polygon into separate pieces, all kept in the one mesh. A hole entirely outside the outline does nothing.

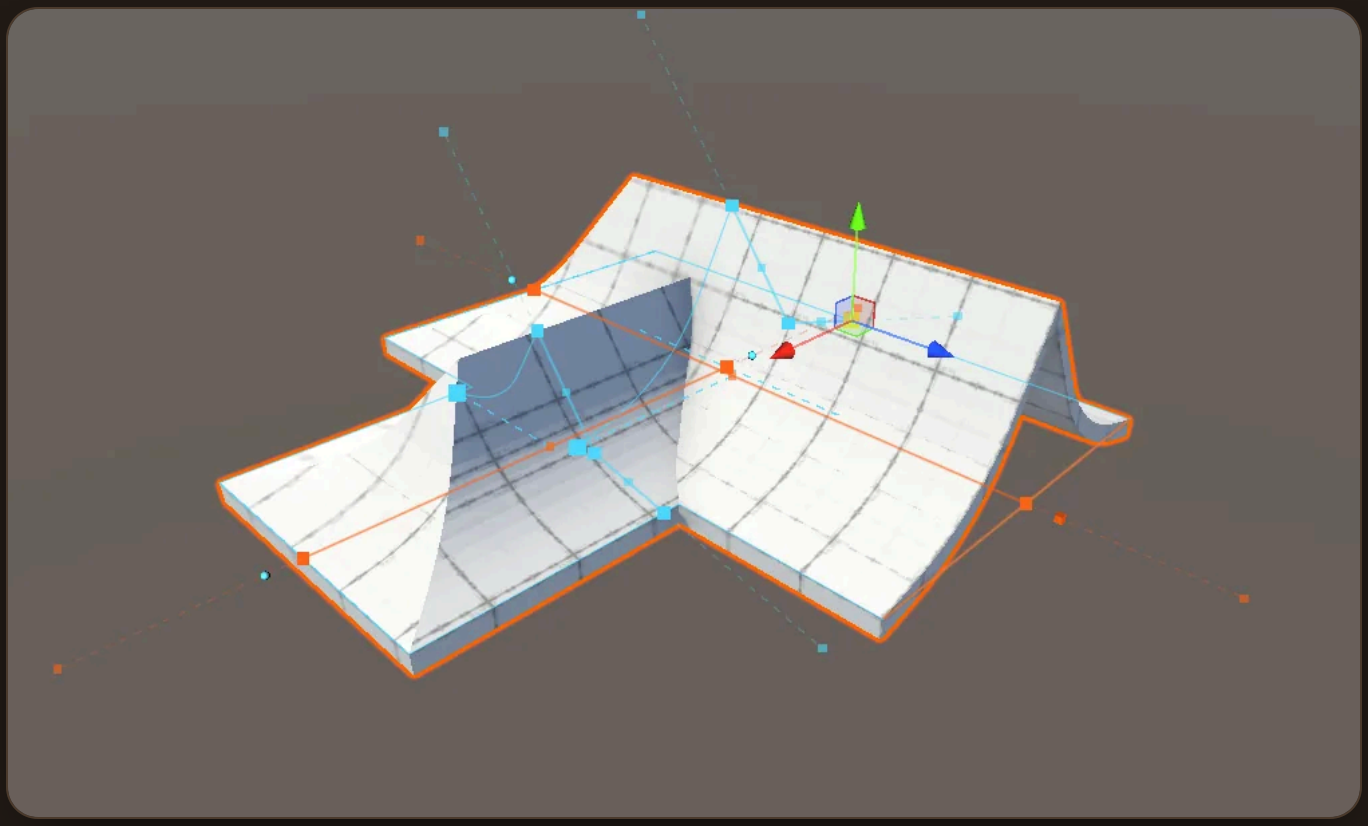
PROPERTY	TYPE	WHAT IT DOES
Stats	Polygon Stats	Shared thickness, subdivision and materials, and the outlines and holes when Save Polygon is on.
Top / Bottom Material	Material	Locked onto the upward and downward facing faces. See Materials.
Thickness	float	How far the polygon is extruded from its outlines. Negative extrudes downwards.
Subdivision Step	float	Cuts the top and bottom into a grid this size, for vertex painting and lighting. Below 0.1 turns it off.
Apply Cuts To ProBuilder Mesh	bool	Make every subdivision cut its own ProBuilder face instead of only splitting the built mesh.
Paths	list	The outlines, each with its own points (on the local XZ plane) and corner rounding.
Holes	list	Holes cut through the polygon, each with its own points and radii.

TIP

Turn on **Save Polygon** on a Polygon Stats asset to store the outlines and holes in it, so every polygon using the asset shares the same shape.

ProBuilder Roof

Hip, valley and gable roofs generated from the center lines of a building's wings. Each wing is a path with its own width and height; wherever wings overlap or touch they are merged into one roof, with hips at the ends, valleys where wings meet and ridges along the middle.



Two wings merged into one roof, with hips, a valley and a gable end.

1. Add `Add Component` › `tsitski ProBuilder Tools` › `tsitski ProBuilder Roof` to an empty GameObject. It starts with a single 8 m wing.
2. Edit each wing's center line in the Scene view (see Editing points and paths). Click **Add Path** to add another wing.
3. Drag the width, height and profile handles, and click the end markers to make gable ends.

Scene view handles

- **Profile** — every wing shows the roof's cross-section on its longest segment. Drag the profile's middle points to reshape it (insert, delete and round them like any path); the shape is shared, so every wing updates.

- **Width** — drag the profile's eave point, or the dot on the opposite side, to change that wing's width. The roof stays symmetric.
- **Height** — drag the profile's ridge point up or down to change that wing's ridge height.
- **Gable ends** — click the marker just past an open end to switch it between a hip (dotted line, sphere) and a vertical gable (solid line, cube).

Profile

The profile runs from the eave at (0, 0) to the ridge at (1, 1) and is stretched to each wing's half width and height. Two points give a plain pitch; more give mansard, gambrel or curved roofs. A flat last segment gives a flat top. Past its own ridge (where a lower wing runs into a taller one) a wing keeps rising at its average pitch, so the two meet in a clean valley.

Thickness, fill and gable walls

By default the roof is a shell like a trim: the top surface is copied straight down by the thickness, with a fascia along the eaves and a band following the profile across each gable end. The underside is left open, so there's room for an attic.

- **Fill Roof** makes it solid instead, with a flat underside at the bottom of the eaves.
- **Generate Gable Walls** builds a wall under each gable, filling the gap between the roof's underside and the bottom of the eaves. **Gable Wall Indent** and **Gable Wall Thickness** set where it sits and how thick it is. Not built on a filled roof.

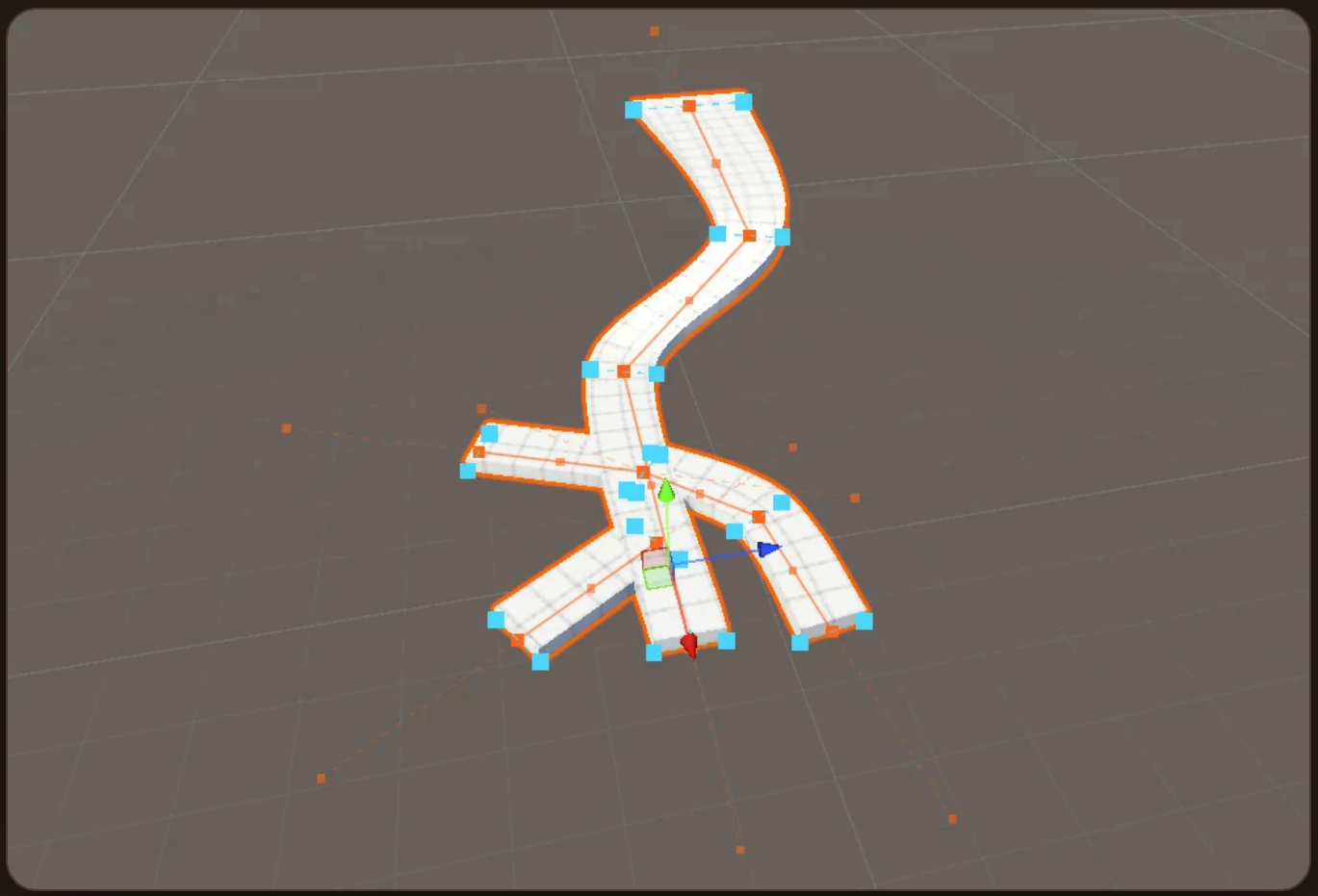
PROPERTY	TYPE	WHAT IT DOES
Stats	Roof Stats	Shared profile, rounding, thickness, fill and gable walls, and the paths when Save Paths is on.
Material	Material	Locked onto every face. See Materials.
Profile	points	The slope's cross-section, from eave (0, 0) to ridge (1, 1).
Profile Radii	floats	Corner rounding per profile point.
Rounding Segments	int	How smooth rounded corners are, on the profile and the paths.
Thickness	float	Thickness of the roof shell.
Smoothing Angle	0–180	Faces meeting at less than this angle are shaded smooth. 0 keeps every edge hard.
Fill Roof	bool	Solid roof with a flat underside instead of a shell.
Generate Gable Walls	bool	A wall under each gable end.
Gable Wall Indent / Thickness	float	How far gable walls sit in from the gable, and how thick they are.
Paths	list	The wings. Each has the settings below.
Width	float	Width of the wing's roof, from eave to eave.
Height	float	Height of the wing's ridge above the eaves.
Closed	bool	Connect the last point back to the first, for a ring around a courtyard.
Gable Start / Gable End	bool	End the wing with a vertical gable instead of a hip.

NOTE

If a wing can't be joined to the rest of the roof cleanly (usually a path that bends back sharply on itself), it's built as its own roof instead and the inspector names it. Nudging its points, width or height usually lets it join again.

ProBuilder Path

A raised strip along curved branches, for sidewalks, kerbs, paths and ledges. Every point has its own width and can sit at its own height, and branches join into junctions wherever they meet.



Branches joined at a T and a crossing.

1. Add `Add Component` › `tsitski ProBuilder Tools` › `tsitski ProBuilder Path` to an empty GameObject, or use `Tools` › `tsitski` › `ProBuilder` › `Create` › `Path`.
2. Edit the branch's points in the Scene view (see Editing points and paths). Points can be at different heights, so a path can climb a slope.

3. Drag the side handles at each point to set its width, and the arrow to set the height.
4. Select a point and click **Branch From Selected Point** to start a new branch there, or click **Add Branch**.

Junctions

- Points of different branches placed on top of each other join into a junction, so branches can meet as a T, a crossing or any number of ways. Dragged points snap onto nearby points of other branches.
- A branch end placed on another branch's curve joins it there too, and dragged ends snap onto other branches' curves.
- Where a branch passes straight through a junction, the widest one stays unbroken and the other branches stop exactly at its edge, so the main path's shape and texture run straight through.
- A branch whose end meets its own start makes a closed loop.

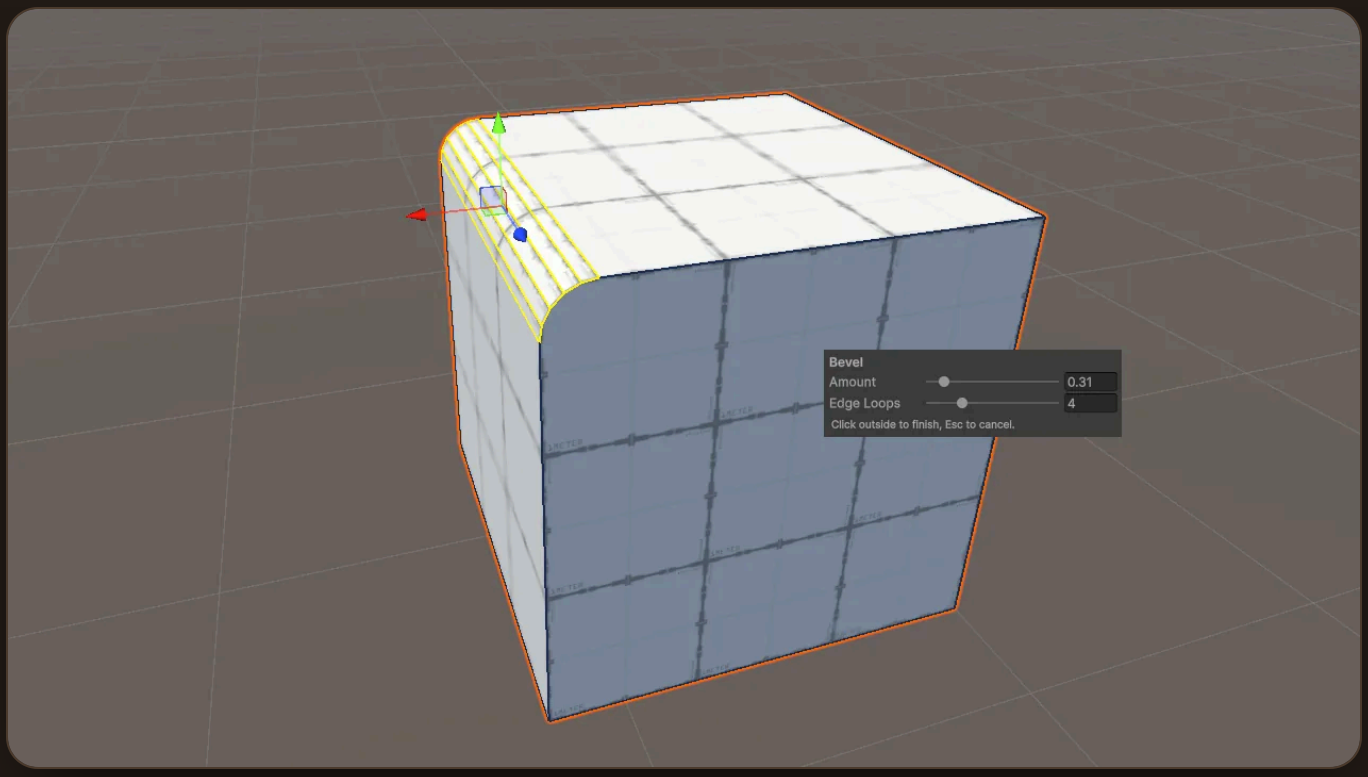
Shape and UVs

- Height only curves between points at different heights, so flat stretches stay flat and slopes stop exactly at their points.
- Wide paths around tight corners pinch on the inside of the turn instead of folding over themselves.
- UVs run along the path: across the width and along its length, in world units, so textures follow the curve.
- The inspector shows a top-down preview of every branch, where points can be moved, added and deleted.

PROPERTY	TYPE	WHAT IT DOES
Stats	Path Stats	Shared sampling, smoothing and materials, the height and width while Store Height and Store Width are on, and the branches when Save Paths is on.
Store Width / Width	bool, float	When on, every point uses this width. Off lets each point keep its own width.
Height	float	How far the path is extruded from its points. Negative extrudes downwards.
Samples Per Segment	int	How many pieces each span between two points is split into along the curve.
Smoothing Angle	0–180	Faces meeting at less than this angle are shaded smooth.
Branches	list	The curves the path follows, each with its points and a width per point.
Top / Side / Bottom Material	Material	Locked onto the top and junctions, the sides and end caps, and the underside. See Materials.

Bevel

Bevels the selected edges in edge mode, or the outer edges of the selected faces in face mode. Run `Tools` › `tsitski` › `ProBuilder` › `Modelling` › `Bevel` or press `Alt` + `B`.

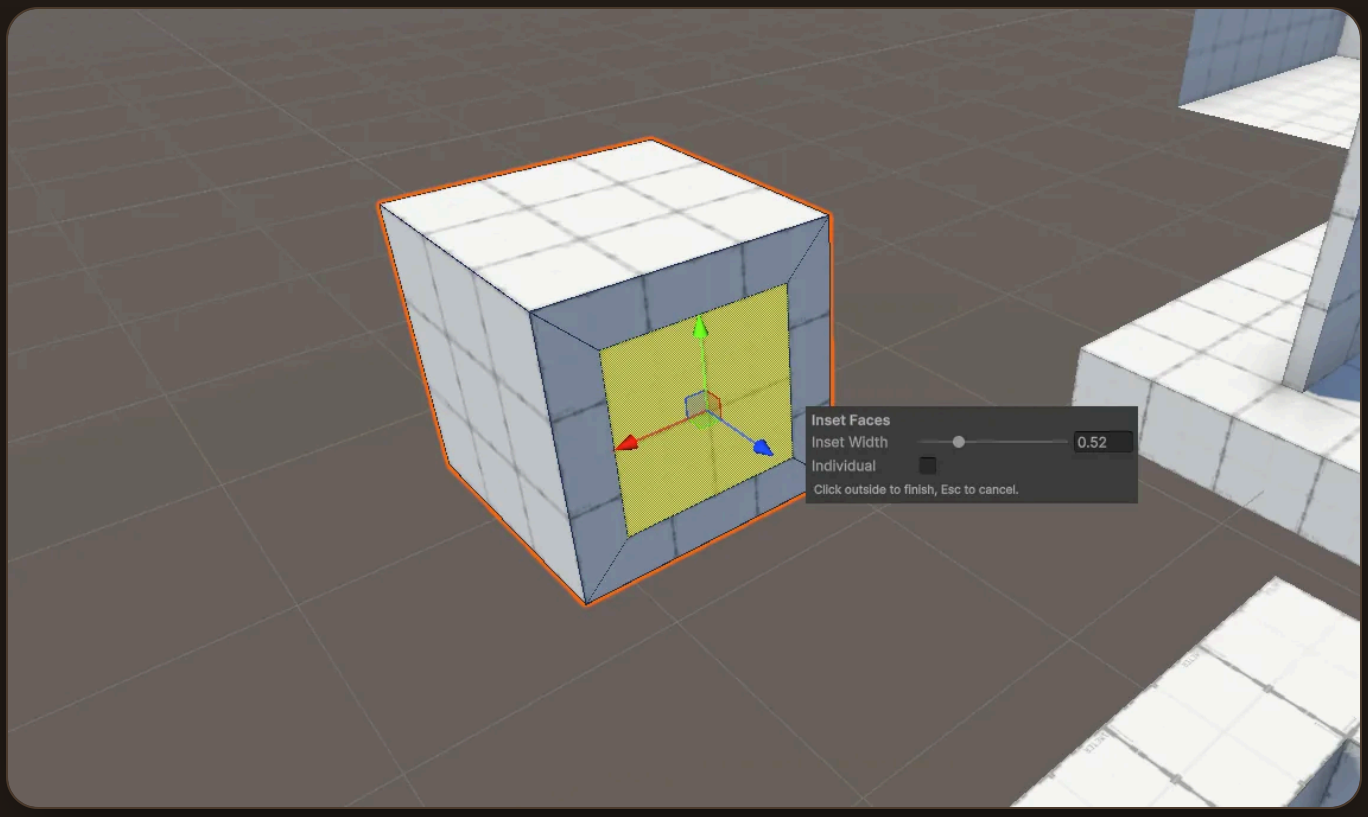


A rounded bevel with Edge Loops.

- **Amount** — how far the bevel reaches. The slider stops at the longest bevel the surrounding edges allow.
- **Edge Loops** — 0 gives a flat chamfer; more loops round the bevel off into true circular arcs, and the faces around it are updated so there are no gaps. Corners where three bevelled edges meet, like a cube's, get a rounded patch made of quads, like Blender's.

Settings window

Bevel and Inset Faces apply straight away and open a small settings window at your cursor, like Blender's redo panel. Drag its sliders and the result updates live.

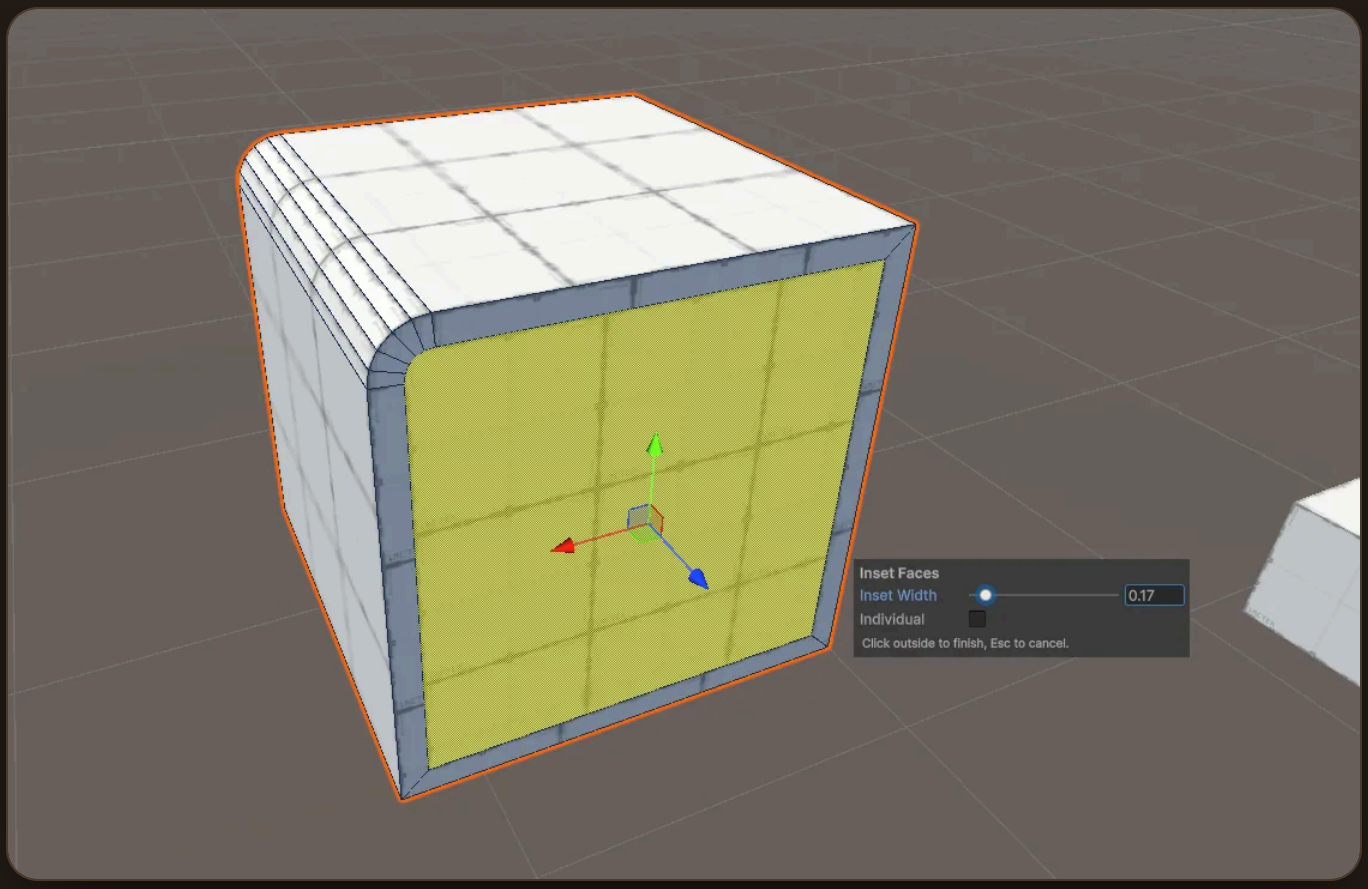


The settings window opens at your cursor.

- Click outside the window, or press `Enter`, to keep the result.
- `Esc` cancels and puts the mesh back.
- The whole operation is one undo step, and the window remembers your last values.

Inset Faces

Insets the selected faces in face mode. Run Tools › tsitski › ProBuilder › Modelling › Inset Faces or press `Alt` + `I`. Like Bevel, it opens a live settings window at your cursor.

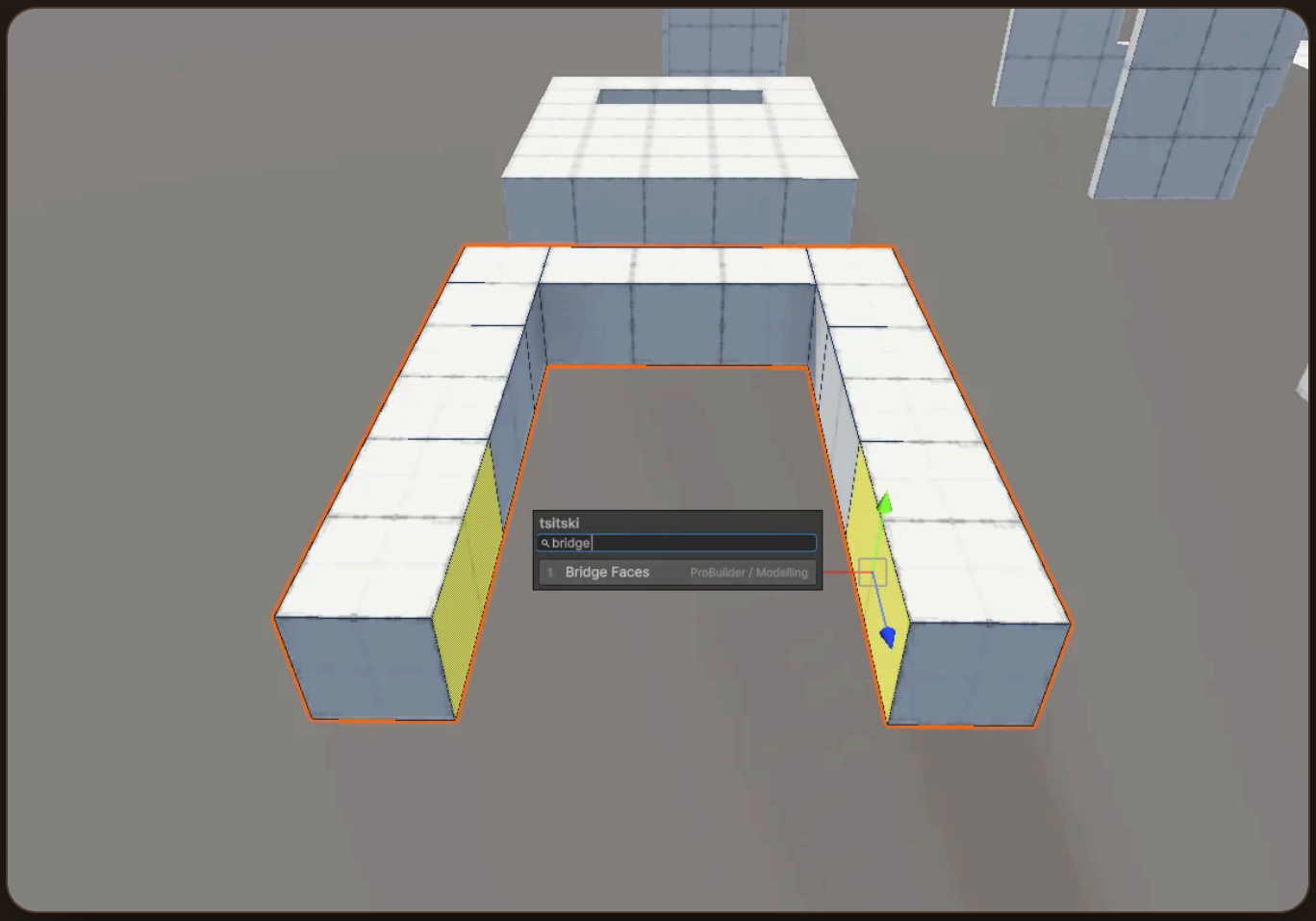


Inset Faces on a group of faces.

- **Inset Width** — how far the new edge sits in from the old one.
- **Individual** — insets each face on its own instead of the selection as one region.

Bridge Faces

Connects two faces, or two groups of faces, with a ring of quads, then deletes the faces you selected. It works like Blender's Bridge Edge Loops, lining the outlines up so the result doesn't twist. Faces that face each other make a tube between them; faces that face away from each other, like the front and back of a wall, make a tunnel through it with its faces pointing inward.



Two faces bridged into a tunnel.

1. Select the mesh and switch ProBuilder to face mode.
2. Select two faces, or two separate groups of faces, on the same mesh.
3. Run Tools › tsitski › ProBuilder › Modelling › Bridge Faces, or pick it from the `Alt` + `X` popup.

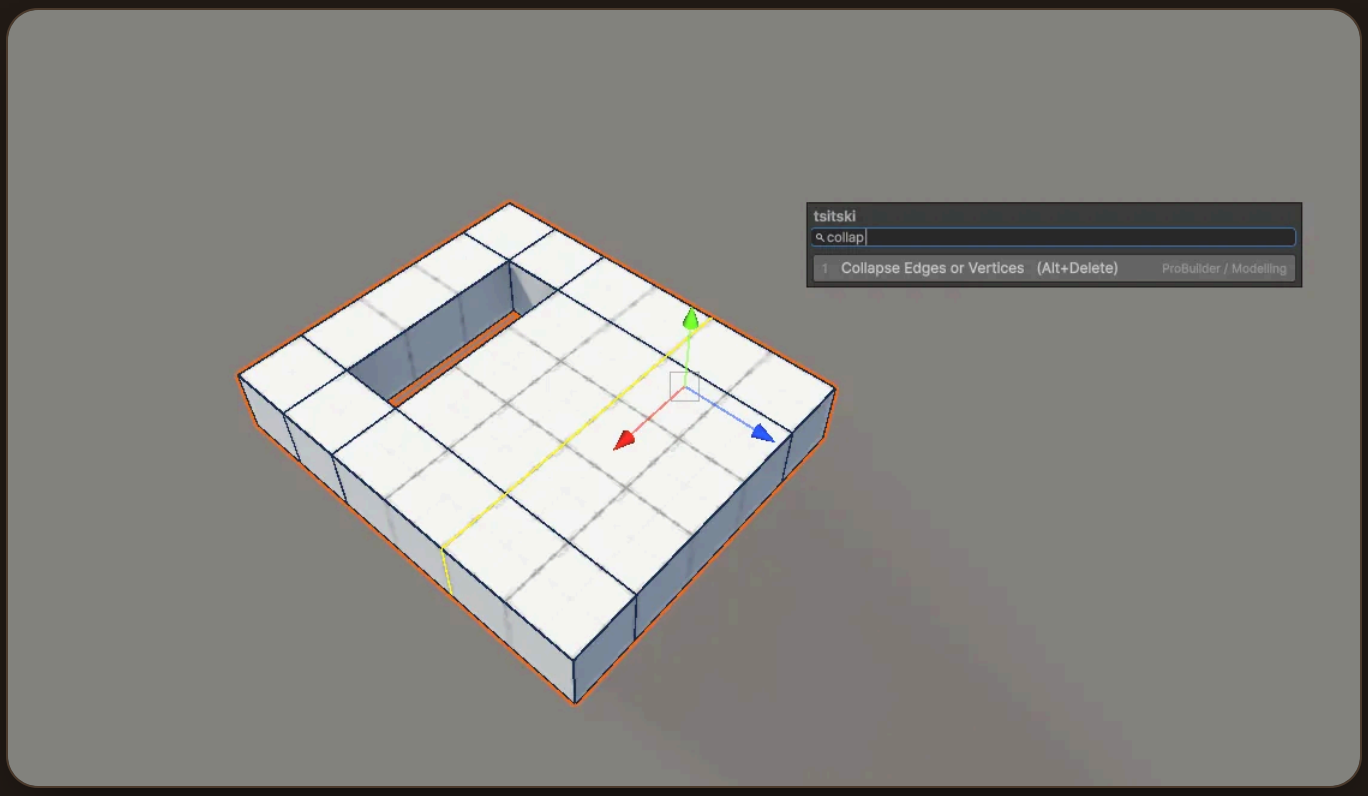
<details>
<summary>What can be bridged</summary>

- Exactly two groups. Faces that share an edge count as one group.
- Each group's outline has to be one loop, with no holes.
- Both outlines need the same number of edges.
- The two groups can't share any vertices.

If something doesn't fit, nothing changes and the Console says why.
</details>

Collapse Edges or Vertices

Cleans up flat geometry. In edge mode it merges the flat faces on both sides of each selected edge. In vertex mode it merges the flat faces around each selected vertex, and in face mode it does the same for every vertex of the selected faces. Vertices left sitting in a straight line along an edge are removed.



Extra edges on a flat wall collapsed into one face.

Press `Alt` + `Delete`, or run
Tools › tsitski › ProBuilder › Modelling › Collapse Edges or Vertices.

> NOTE
> Only faces that lie flat against each other are merged. Corners are skipped, and the Console says which.

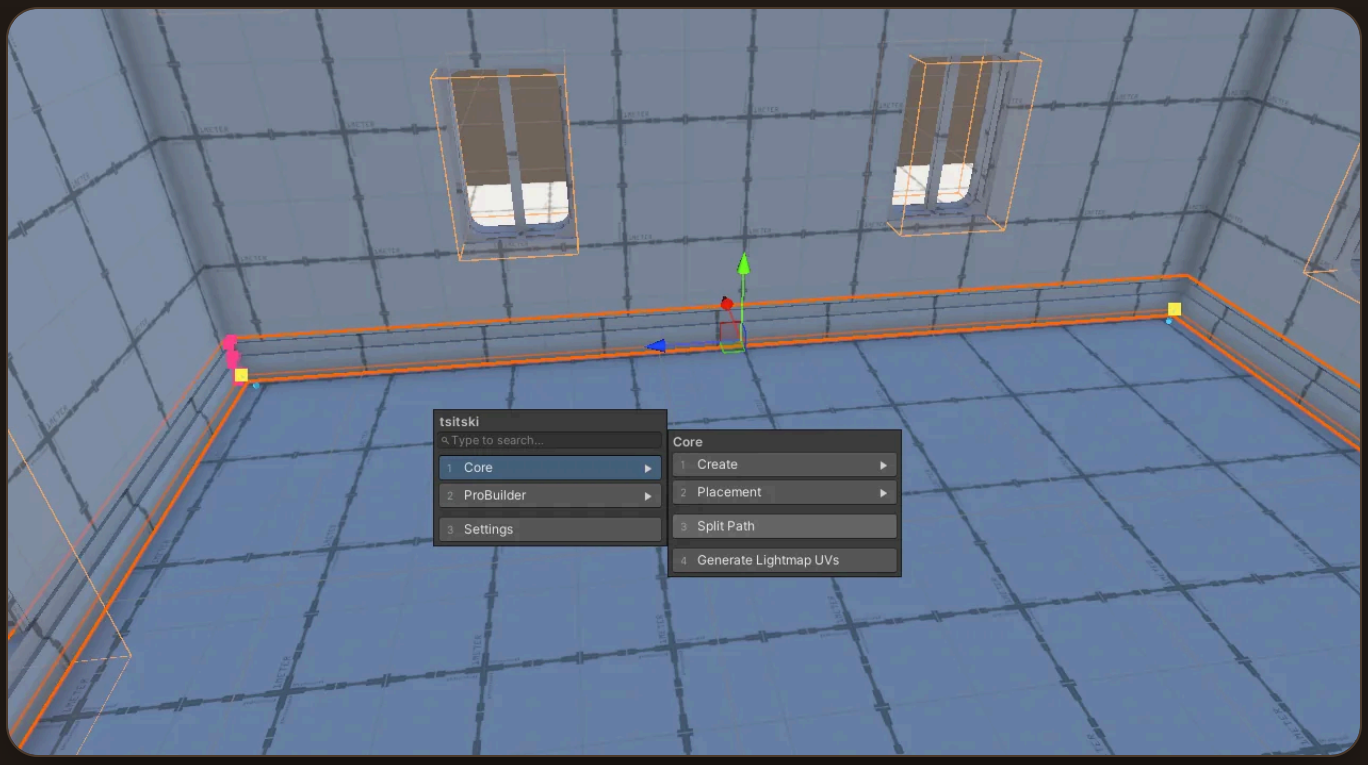
Placement

Quick ways to put selected objects where they belong, under
Tools › tsitski › Core › Placement. Each one is a single undo step.

PROPERTY	TOOL	WHAT IT DOES
Snap To Grid	Placement	Round each position to the grid set in the Scene view's snap settings.
Snap To Oriented Grid	Placement	Snap to a grid turned to the rotation most of the selection shares, and round rotations to whole degrees, so rotated rows stay lined up.
Snap To Oriented Grid And Angle	Placement	Like Snap To Oriented Grid, and also round rotations to the Scene view's rotation snap.
Snap To Wall	Placement	Move each object onto the nearest surface along its own axes and turn it to face the surface. Only snaps onto the Placement Layers in Settings.
Step Physics	Placement	Drop the selected objects with physics for a second and leave them where they settle, so props land naturally on shelves and floors.

Split Path

Select path points on a trim, array, stairs, polygon, roof, floor plan, path or cutter (or a fence, mesh decal, path assembly or support with tsitski Mesh Tools) and run `Tools` › `tsitski` › `Core` › `Split Path`. Every stretch between neighbouring selected points becomes a new object with the same settings, and the pieces stay joined. Per-point settings like widths and corner radii carry over, and each piece gets its own mesh. The new objects are selected, and the split is one undo step.



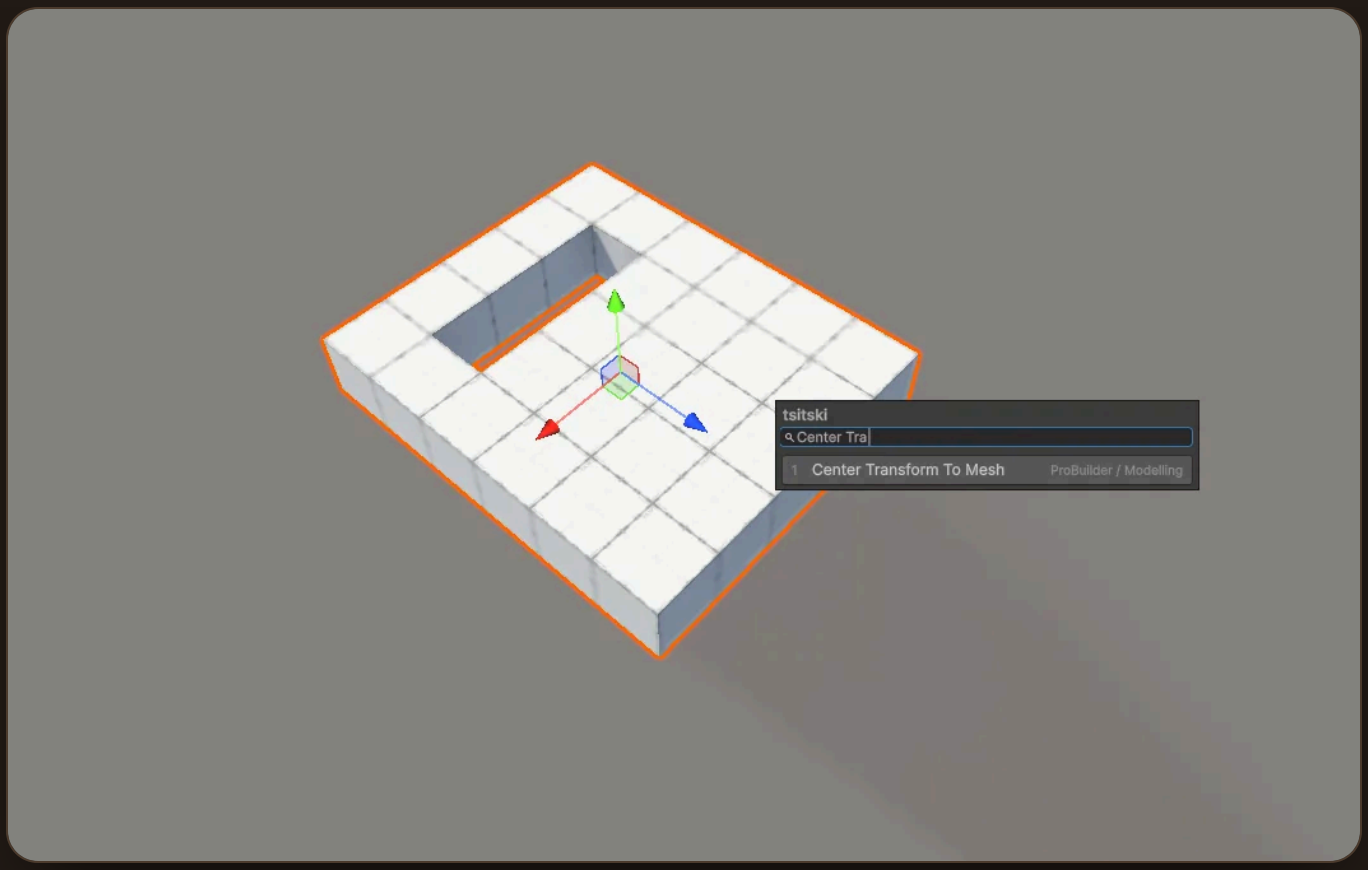
The selected stretch becomes its own trim.

- On a polygon, select a run of at least three outline points to cut it in two; each piece becomes its own closed outline.
- On objects with several paths, select every point of one path to move that whole path into a new object.

> NOTE
> Paths stored in a Stats asset can't be split, since other objects share them. Spiral stairs have no path to split.

Center Transform To Mesh

Moves each selected object's pivot to the centre of its mesh, snapped to the grid. The mesh and its children stay exactly where they are; only the pivot moves. Generators move all of their own points (paths, holes, branches and wings) so they rebuild in the same place.



The pivot moves to the middle; the mesh stays put.

Select one or more objects and run `Tools` › `tsitski` › `Core` › `Center Transform To Mesh`. It works on plain ProBuilder and mesh objects and on every generator: polygons, roofs, paths, stairs, trims, floor plans and arrays (and fences, mesh decals and supports with tsitski Mesh Tools).

> NOTE
> Objects that can't move their pivot are skipped with a Console message: paths stored in a Stats asset, pieces generated by another object, pillars and spiral stairs (built around their pivot), and circle or rectangle mesh decals.

Generate Lightmap UVs

Generators make their own lightmap UVs when they rebuild. To redo them all at once, for example before a final bake, run `Tools` › `tsitski` › `Core` › `Generate Lightmap UVs`. It covers every generated and ProBuilder mesh in the open scenes that contributes to global

illumination. To redo a single mesh, right-click its Mesh Filter and choose **Generate Lightmap UVs**.

Setup Materials

Puts materials on every ProBuilder mesh under the selected objects in one go. Select a building (or several) and run `Tools` › `tsitski` › `ProBuilder` › `Setup Materials`.

- **Simple** puts one material on every face.
- **Complex** sorts faces into slots by casting rays to see whether each one is enclosed: **Inside / Outside Wall**, **Inside / Outside Floor**, **Inside / Outside Roof** (ceilings and the underside of roofs count as inside roof), and **Inside / Outside Trim**. **Inside Distance** sets how far each face looks for other geometry.
- Whole trims, arrays and pillars count as one piece, inside or outside, and use the trim slots. Empty trim slots fall back to the wall materials.
- Empty slots leave those faces as they are.
- Meshes whose materials are locked by their generator are skipped (see Materials).

The window remembers its materials. **Save As** stores them in a **Building Material Group** asset, which you can pick in the **Group** field to reuse them on other buildings. With Real Blend installed, every slot also picks the Real Blend layer to paint those faces with. The whole setup is one undo step.

Outline Trim

Select a mesh, a floor plan or a floor plan's floor and run `Tools` › `tsitski` › `ProBuilder` › `Generate` › `Outline Trim` to generate closed trims or arrays around its outline. The settings window previews the result live, and the whole thing is one undo step.

- **Create** — a Trim, or an Array of meshes along the outline. Each uses the optional Trim Stats or Array Stats you pick.
- **Side** — Outside, Inside or Center of the walls. Inside on a mesh with no inner outline runs inward along the edge.

- **Height** — Bottom or Top of the mesh, with an **Offset** away from the wall surface.

Snap Onto Path

Select objects and run `Tools` › `tsitski` › `ProBuilder` › `Snap Onto Path` to move each one onto the nearest floor plan, trim, array or stairs path and turn it to follow the path.

- **Offset** — moves the objects sideways off the path.
- **Floor Plan Center** — only snap onto floor plans, centred inside the thickness of their walls.
- **Snap Height** — move the objects to the path's height, instead of keeping their own.

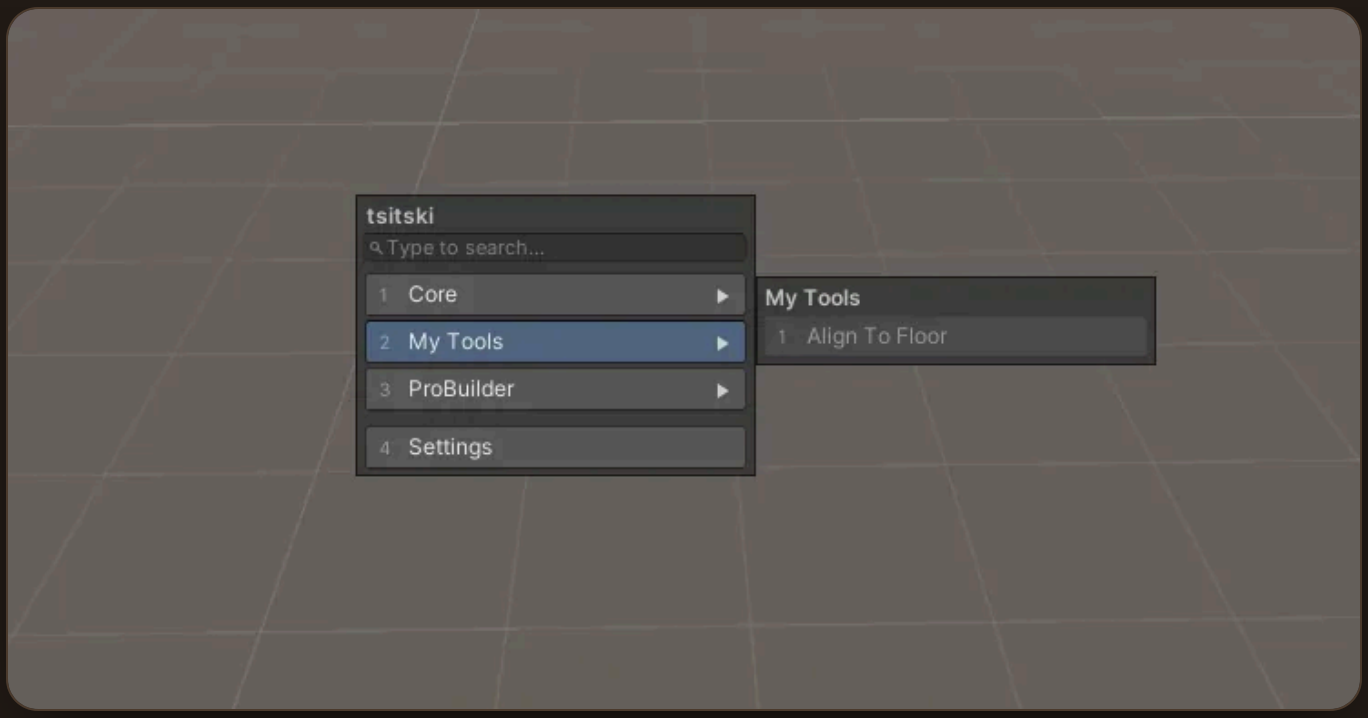
Adding your own tools

Your own editor tools can show up in the tools popup and its search. Put the `[TsitskiMenu]` attribute on a static method with no parameters. The path picks the group and name, slashes open submenus.

```
using Tsitski.Core;
using UnityEditor;

public static class MyTools {
    [TsitskiMenu("My Tools/Align To Floor", validate = nameof(HasSelection), priority
    private static void AlignToFloor() {
        // ...
    }

    private static bool HasSelection() => Selection.transforms.Length > 0;
}
```



Your tool, in its own group in the popup.

PROPERTY	TYPE	WHAT IT DOES
path	string	Group and name in the popup, like "My Tools/Align To Floor".
validate	string	Name of a static bool method in the same class. The entry is greyed out when it returns false.
priority	int	Sort order. Lower comes first, and a gap of more than 10 adds a separator. Default 1000.
shortcut	string	Shortcut text shown next to the name.

TIP

A regular `[MenuItem("Tools/tsitski/...")]` shows up in the popup too, with its validate method and shortcut.

Settings

Tools › tsitski › Settings opens the settings, stored in `tsitski Settings.asset` and shared with other tsitski packages. It's created in the `tsitski Core` folder the first time it's needed,

and can be moved anywhere in the project.

PROPERTY	TYPE	WHAT IT DOES
Default Mesh Material	Material	Given to generated meshes (polygons, roofs, trims, arrays, stairs, pillars and walls) that don't have one yet.
Rebuild After Dragging	bool	Wait until a dragged handle is released before recutting and rebuilding. Turn off to rebuild live while dragging.
Auto Name Generators	bool	Name generator objects after their Stats asset, or after the component when they have none. Turning it on renames everything already in the scene.
Gizmos Only On Selected	bool	Only draw cutter gizmos for selected cutters.
Contribute GI By Default	bool	Turn on the Contribute GI static flag on objects when a generator is added, and on the child objects generators create.
Placement Layers	LayerMask	Layers Snap To Wall snaps onto. See Placement.
Show Review Button	bool	Show the Leave A Review button at the bottom of the tools popup.

NOTE

Generators only rebuild when their settings actually change, so opening a scene, reloading scripts, leaving play mode or undoing something unrelated doesn't rebuild them or reset their lightmap UVs.

Troubleshooting

▾ My cutter isn't cutting anything

- The mesh has to be active and in the same scene as the cutter, and a ProBuilder mesh unless **Cut Mode** includes regular meshes.

- The cutter has to actually go into the mesh, not just touch its surface.
- Every axis of the cutter's Size has to be bigger than 0.
- A cutter never cuts meshes in its own prefab or among its own children.
- A cutter that's a child of a ProBuilder mesh skips that mesh unless **Cut Through Parent** is on.
- Still stuck? Run `Tools` › `tsitski` › `ProBuilder` › `Generate` › `Rebuild All Cutters`.

The inside walls are missing or doubled

Check **Create Mesh** and **Generate Faces** on the cutter. When one cutter goes through several meshes, only one of them gets the walls; pick which with **Faces On**.

A roof wing is built on its own, or part of a roof is missing

When a wing can't be joined to the rest of the roof cleanly it's built as its own roof, and a part that can't be built at all is left out; the roof's inspector says which. This usually comes from a path that bends back sharply on itself. Nudge its points, width or height, or split it into two paths.

My edits to a polygon, roof, trim, array, stairs or pillar mesh disappeared

Generated meshes are rebuilt from their settings, so direct ProBuilder edits to their shape are overwritten. Change the component's settings, path or profile instead. Face materials and manually edited UVs are kept.

A floor plan's trim, array or roof I deleted came back, or won't come back

Deleted generated parts stay deleted and are listed under **Removed Layers** on the floor plan. Click **Restore** to bring one back.

▾ Bridge Faces says it can't bridge my faces

The Console message names the problem. See What can be bridged.

Credits

The cutter's boolean geometry is built on CSG code by Evan Wallace (csg.js), Tomasz Dabrowski (csgjs-cpp) and Karl Henkel (C# port), used under the MIT licence. The full notice ships with the package in `Third-Party Notices.txt`.

Questions, bugs or ideas? Come say hi on Discord.